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Complex dynamics of epidemic models on adaptive networks ☆

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Abstract

There has been a substantial amount of well mixing epidemic models devoted to characterizing the observed complex phenomena (such as bistability, hysteresis, oscillations, etc.) during the transmission of many infectious diseases. A comprehensive explanation of these phenomena by epidemic models on complex networks is still lacking. In this paper we study epidemic dynamics in an adaptive network proposed by Gross et al., where the susceptibles are able to avoid contact with the infectious by rewiring their network connections. Such rewiring of the local connections changes the topology of the network, and inevitably has a profound effect on the transmission of the disease, which in turn influences the rewiring process. We rigorously prove that the adaptive epidemic model investigated in this paper exhibits degenerate Hopf bifurcation, homoclinic bifurcation and Bogdanov–Takens bifurcation. Our study shows that adaptive behaviors during an epidemic may induce complex dynamics of disease transmission, including bistability, transient and sustained oscillations, which contrast sharply to the dynamics of classical network models. Our results yield deeper insights into the interplay between topology of networks and the dynamics of disease transmission on networks.

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1. Introduction

Mathematical models have been always at the core of understanding and revealing the transmission dynamics of infectious diseases since Bernoulli investigated the spread of smallpox in 1760 [2]. Traditionally, mathematical epidemic models are formulated by differential equations, and classical compartmental models operate under an assumption of a random and homogeneous mixing population, where each individual is treated similarly and indistinguishably from the others, and has an equal chance of contacting with any other individuals [1,9]. In reality, however, each individual has his or her own social network on which epidemic transmits, and the number of contacts for each individual is considerably smaller than the population size. Hence, the assumption of random and homogeneous mixing population may be too idealized, and it is reasonable to model infectious diseases incorporating complex social network structure [15,16,19,20,28]. For this reason, modeling infectious diseases on complex networks has attracted tremendous interest over recent years. By representing a population and connections between individuals as a network on which the nodes represent individuals and links represent various interactions among these individuals, the infectious diseases in a population are considered to spread through a network of the complex interactions between individuals or groups, and changes in the status of individuals depend on the status of their neighbors [5,19,21–23].

An impressive research effort has been devoted to understanding the effects of complex network topologies on disease transmission by mathematical models. Pastor-Satorras et al. proposed a heterogeneous mean-field approach and established the first SIS model on complex networks to analyze the epidemic spreading [21,22]. Their approach assumes that all nodes with the same degree are statistically equivalent, and the population is divided into different compartments depending on the states and degrees, such as the susceptible and infectious with degree k (denoted by S_k and I_k, respectively). The epidemic threshold $\lambda_c = \langle k\rangle/\langle k^2\rangle$ is obtained on uncorrelated complex networks, where $\langle k\rangle$ and $\langle k^2\rangle$ are the first-order (i.e., average degree) and second-order moments of degree distribution of the network, respectively. Later on, Wang et al. analyzed the global dynamics of this model, and proved that if the effective transmission rate is less than the epidemic threshold λ_c, the disease will die out; otherwise, the unique endemic equilibrium is globally asymptotically stable [31]. It is worth pointing out that, this result recovers the epidemic threshold $\lambda_c = 1/\langle k\rangle$ on a complete homogeneous (regular) network due to the fact that $\langle k^2\rangle = \langle k\rangle^2$. Since then this framework of general methodology has been extended to a wealth of dynamical models on networks, where topological features of complex networks were explicitly addressed, for instance, the degree correlations [5,6], heterogeneity [10] and asymmetry [18,33], etc. However, most works ignored the change of network topology during the disease transmission, and only considered the static networks, on which the network topology remains fixed during an epidemic. It has been shown in these works that there are no oscillations, and the disease transmission dynamics were completely determined by the basic reproduction number R_0, i.e., the disease will disappear and persist when $R_0 < 1$ and $R_0 > 1$, respectively.

Typically, the classical homogeneous compartment epidemic models have been proven to be capable of supporting rich dynamics. It turns out that some kinds of ingredients can lead to complex dynamics including bistability and oscillations, such as nonlinear incidence rate [17,24], nonlinear recovery rate [25,26,32] and antibiotic resistance [7], etc. However, there are limited research of epidemic models on networks showing these complex dynamics. In particular it is not clear which factors will lead to rich dynamics like multistability, recurrence of epidemics, etc.

In order to incorporate the change of network topology into the disease transmission, Keeling et al. proposed a network-based pairwise epidemic model, and introduced the variable $[AB]$, i.e., the number of pairs $A-B$ of an individual in state A in contact with another individual in state B in the network, where the pair $A-A$ is calculated twice because of symmetry. The quantity of $[AB]$ counts local spatial connections, and can therefore describe the local topological structures of the network. In order to study the evolution of $[AB]$ over time, they introduced the variable $[ABC]$, which represents the number of triple $A-B-C$ of an individual with state A connected to an individual with state B connected to an individual with state C in the network [13,15]. Considering that humans tend to protect themselves by avoiding contacts with the infectious, Gross et al. [11] extended the SIS pairwise epidemic model to adaptive networks as follows

$$\begin{cases} [\dot{S}] = \gamma[I] - \tau[SI], \\ [\dot{I}] = -\gamma[I] + \tau[SI], \\ [\dot{SI}] = \tau[SSI] - (\tau+\gamma)[SI] + \gamma[II] - \tau[ISI] - w[SI], \\ [\dot{SS}] = 2\gamma[SI] - 2\tau[SSI] + 2w[SI], \\ [\dot{II}] = 2\tau[SI] + 2\tau[ISI] - 2\gamma[II], \end{cases} \tag{1.1}$$

where $[S]$ and $[I]$ denote the total numbers of susceptible and infectious individuals, respectively. Infectious individuals can infect each of their susceptible neighbors with a rate τ and recover with a rate γ per unit of time. The susceptibles can disconnect the link with an infectious individual with a rate w and rewire that link to a randomly chosen susceptible individual. Notice that model (1.1) is not in closed form, one needs to express $[ABC]$ with the lower forms of connections. Therefore, a moment closure approximation formula for the triple $[ABC]$ in terms of pairs and singles was proposed by Keeling et al. [13,14] under the assumption of the degree distribution subject to a Poisson distribution, which is

$$[ABC] \approx \frac{[AB][BC]}{[B]},$$

where $A, B, C \in \{S, I\}$. Then Gross et al. [11] studied the closed form of model (1.1) and investigated how rewiring process changes dynamics of disease transmission numerically. They showed that adaptive networks can give rise to rich dynamics such as oscillations, hysteresis and bistability. Similar dynamical behaviors were also found in an SIRS epidemic model on the adaptive networks by Shaw et al. [27]. All these works indicated that changes of network topology can affect the transmission of disease, and lead to complex dynamics that are absent in epidemic models on static networks.

Nevertheless, the results mentioned above are numerical [11,27], and people still can not fully understand the connections between rewire process and disease transmission. Surprisingly, there is very few theoretical work towards this direction. Recently Bodó et al. [3] gave a characterization of existence and stability of equilibria of the SIS epidemic model on the adaptive network, and showed that the Hopf bifurcation can be present. In this paper we will provide a detailed analysis of the model (1.1). We firstly reduce the closed form of model (1.1) to a three dimensional system by some reasonable assumptions, and then we analyze the backward bifurcation, saddle-node bifurcation, Hopf bifurcation and Bogdanov–Takens bifurcation of codimension 2. We would like to emphasis that our theoretical analysis not only explains the numerical findings in [3,11,27], but also helps to discover a new phenomenon of two limit cycles, that was never found before. Our study implies that Bogdanov–Takens bifurcation of cod-2 and Hopf bifurcation of at least cod-2 are two different mechanisms to generate oscillating dynamics in adaptive-network epidemic models.

The paper is organized as follows. In Section 2, we derive an equivalent system to the closed form of model (1.1) by assumptions and changes of variables. Section 3 is devoted to the study of the existence and stabilities of equilibrium points, and the proofs of backward bifurcation and saddle-node bifurcation are given. In Section 4, we give detailed analysis of Hopf bifurcation and Bogdanov–Takens bifurcation of codimension 2. Numerical simulations illustrating the theoretical results are also provided. We summarize our results and give a brief discussion in Section 5.

2. SIS pairwise epidemic model

The epidemic model (1.1) is based on the bidirectional connected adaptive network $\mathcal{G} = (V, L)$, where V is the set of nodes and L is the set of edges. Let $N = |V|$ and $K = |L|$ denote the constant numbers of nodes and bidirectional links of the network $\mathcal{G}$, respectively. Then the average degree of the network can be represented as $\langle k\rangle = \frac{2K}{N}$. Assume that there are no self-loops and multiple connections between nodes in a rewiring process. Since the network is connected, there are no isolated nodes and it is not difficult to verify that $\langle k\rangle > 1$ when $N > 2$.

Notice that $|V| = N$, and $[SS]$ and $[II]$ count twice for the corresponding pairs, we have two balanced relationships $[S] + [I] = N$ and $[SS] + 2[SI] + [II] = 2K$. By normalization

$$s = \frac{[S]}{N}, \quad i = \frac{[I]}{N}, \quad P_{SS} = \frac{[SS]}{2K}, \quad P_{SI} = \frac{[SI]}{K}, \quad P_{II} = \frac{[II]}{2K},$$

we have $s + i = 1$ and $P_{SS} + P_{SI} + P_{II} = 1$, and the closed form of model (1.1) reduces to

$$\begin{cases} \dot{i} = -\gamma i + \frac{\tau\langle k\rangle}{2} P_{SI}, \\ \dot{P_{SI}} = \frac{\tau\langle k\rangle}{2(1-i)} P_{SI}(2P_{SS} - P_{SI}) - (\tau + 3\gamma + w)P_{SI} - 2\gamma P_{SS} + 2\gamma, \\ \dot{P_{SS}} = P_{SI}\left(\gamma + w - \tau\langle k\rangle \frac{P_{SS}}{1-i}\right), \end{cases} \tag{2.1}$$

where the derivatives are taken with respect to time $\tilde{t}$.

For biological feasibility, we only consider system (2.1) in the domain

$$\Omega_0 = \left\{(i, P_{SI}, P_{SS}) \in R^3_+ \,\middle|\, 0 \leqslant i < 1, 0 \leqslant P_{SI} + P_{SS} \leqslant 1\right\}.$$

It is easy to verify that Ω_0 is a positive invariant set of system (2.1).

Model (2.1) includes four parameters γ, τ, ω and $\langle k\rangle$. We further simplify the model by changes of variables and parameters

$$x=\frac{1}{1-i}, \quad y=P_{SI}, \quad z=P_{SS}, \quad t=\gamma\tilde{t},$$
$$\alpha=\frac{\tau}{\gamma}+2, \quad \beta=\frac{\tau\langle k\rangle}{2\gamma}, \quad \mu=\frac{w}{\gamma}+1,$$

where $\alpha>2$, $\beta>0$ and $\mu>1$, and then system (2.1) becomes

$$\begin{cases}\dot{x}=x(1-x+\beta xy),\\ \dot{y}=2(1-z)-(\alpha+\mu)y+\beta xy(2z-y),\\ \dot{z}=y(\mu-2\beta xz),\end{cases} \tag{2.2}$$

where the derivatives are taken with respect to time t.

Since $\langle k\rangle>1$, so $\alpha<2(1+\beta)$ always holds. For convenience, define the parameter space

$$\Lambda=\{(\alpha,\beta,\mu)|\beta>0,\ \mu>1,\ 2<\alpha<2(1+\beta)\}.$$

Recall that $i\in[0,1)$ is the fraction of infectious individuals in a population, the variable x in system (2.2) is greater than or equal to 1. Due to the invariance of Ω_0, we have

$$\Omega=\left\{(x,y,z)\in R_+^3|\,x\geqslant 1,\ y,z\geqslant 0,\ 0\leqslant y+z\leqslant 1\right\}$$

is positive invariant with respect to the flow of system (2.2).

In the rest of the paper, we will study the dynamics and bifurcations of system (2.2) in Ω with parameters in Λ.

3. Equilibrium points

3.1. Existence of equilibrium points

In this section we will study the existence of equilibrium points.

It is obvious that system (2.2) always has a unique disease-free equilibrium $E_0(1,0,1)$. By the next generation matrix method [8,30], we calculate the basic reproduction number

$$R_0=\frac{2\beta}{\mu}.$$

If system (2.2) has an endemic equilibrium $E(x,y,z)$, it must satisfy the following relations

$$y(x)=\frac{x-1}{\beta x}, \quad z(x)=\frac{\mu}{2\beta x}, \quad x>1,$$

where x is a root of the quadratic polynomial

$$F(x)=(x-1)^2+(\alpha-2\beta)(x-1)+\mu-2\beta.$$

Equation $F(x)=0$ has two real roots

$$x_1=\frac{2+2\beta-\alpha+\sqrt{\Delta}}{2} \quad \text{and} \quad x_2=\frac{2+2\beta-\alpha-\sqrt{\Delta}}{2},$$

if its discriminant

$$\Delta=(\alpha-2\beta)^2+4(2\beta-\mu)$$

is positive. We denote two possible endemic equilibria by $E_i\left(x_i,\frac{x_i-1}{\beta x_i},\frac{\mu}{2\beta x_i}\right)$ $(i=1,2)$.

For $(\alpha,\beta,\mu)\in\Lambda$, we have the following three cases about the existence of endemic equilibria.

(1) $R_0>1$. In this case $2\beta>\mu$ and $\Delta>0$. Then we have $x_1>1$ and $x_2<1$. Hence, system (2.2) has a unique endemic equilibrium $E_1\,(x_1,y(x_1),z(x_1))$.
(2) $R_0=1$. In this case $2\beta=\mu$. Clearly, equation $F(x)$=0 has two real roots, 1 and $2\beta-\alpha+1$. Since $\alpha>2$, system (2.2) has a unique endemic equilibrium $E_1\big(x_1,y(x_1),z(x_1)\big)$ if and only if $2<\alpha<2\beta$.
(3) $R_0<1$. In this case $2\beta<\mu$. We have two subcases.
  (i) If $\alpha\geqslant 2\beta$, then $F(x)>0$ for all $x>1$. Therefore, system (2.2) has no endemic equilibrium.
  (ii) If $\alpha<2\beta$, $F(x)=0$ has two roots $x_i>1$ $(i=1,2)$ if and only if $\Delta>0$. $F(x)=0$ has one root of multiplicity 2, $x_1=x_2>1$, if and only if $\Delta=0$. Therefore, system (2.2) has two endemic equilibria $E_1\big(x_1,y(x_1),z(x_1)\big)$ and $E_2\big(x_2,y(x_2),z(x_2)\big)$ if and only if $\Delta>0$, and has a unique endemic equilibrium of multiplicity 2, $E_1=E_2$, if and only if $\Delta=0$.

From what has been discussed above, we have the following results.

Theorem 3.1. *For system (2.2) with $(\alpha,\beta,\mu)\in\Lambda$ and $(x,y,z)\in\Omega$,*

(1) *The disease-free equilibrium E_0 always exists.*
(2) *When $R_0>1$, there exists a unique endemic equilibrium E_1.*
(3) *When $R_0=1$, there exists a unique endemic equilibrium E_1 if and only if $2<\alpha<2\beta$.*
(4) *When $R_0<1$, and*
  (a) *If $\alpha\geqslant 2\beta$, there is no endemic equilibrium.*
  (b) *If $2<\alpha<2\beta$, system (2.2) has two endemic equilibria E_1 and E_2 if and only if $\Delta>0$, and these two equilibria coalesce into one endemic equilibrium $E_1=E_2$ if and only if $\Delta=0$. Otherwise, system (2.2) has no endemic equilibrium.*

In the following, we explore conditions for the existence of equilibria in the parameter set Λ. In particular, we will sketch the bifurcation diagrams in (α,μ) plane with different values of β. For fixed β, the feasible region for (α,μ) is $D=\{(\alpha,\mu)|2<\alpha<2(1+\beta),\mu>1\}$.

If $(\alpha,\beta,\mu)\in\Lambda$ and $0<\beta\leqslant\frac{1}{2}$, then $R_0<1$ and $\alpha>2\beta$ always hold for any $(\alpha,\mu)\in D$. By Theorem 3.1, system (2.2) has no endemic equilibrium. We denote D as D_0 in this case. See Fig. 1(a).

If $(\alpha,\beta,\mu)\in\Lambda$ and $\frac{1}{2}<\beta\leqslant 1$, then $\alpha>2\beta$ is always true. If $\mu\geqslant 2\beta$, i.e., $R_0\leqslant 1$, there is no endemic equilibrium; if $\mu<2\beta$, i.e. $R_0>1$, there exists a unique endemic equilibrium. In

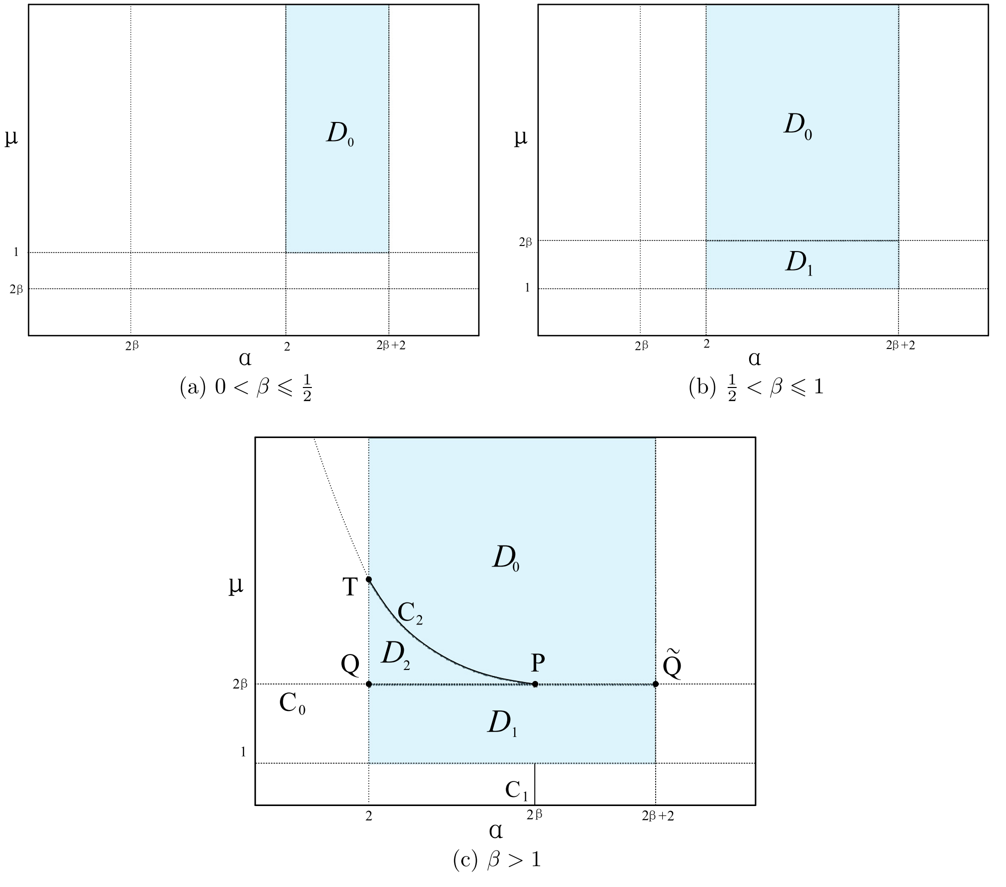


Fig. 1. Bifurcation diagrams of equilibrium points for different values of β. Disease-free equilibrium E_0 always exists in region D. The subscript i of D_i $(i = 0, 1, 2)$ denotes the number of endemic equilibrium points in region D_i. The basic reproduction number R_0 is less than 1 if $(\alpha, \mu) \in D_0 \cup D_2$, and R_0 is greater than 1 if $(\alpha, \mu) \in D_1$. $R_0 = 1$ if (α, μ) lies on the line segment $Q\tilde{Q}$.

this case, region D is divided into two subregions D_0 and D_1 by the threshold condition $\mu = 2\beta$. See Fig. 1(b).

If $(\alpha, \beta, \mu) \in \Lambda$ and $\beta > 1$, as shown in Fig. 1(c), define subregions of D:

$$
\begin{aligned}
D_0 &= \{(\alpha, \mu) | \{2 < \alpha < 2\beta, \mu > \tfrac{1}{4}(\alpha - 2\beta)^2 + 2\beta\} \cup \{2\beta \leqslant \alpha < 2(1+\beta), \mu > 2\beta\}\}, \\
D_1 &= \{(\alpha, \mu) | 2 < \alpha < 2(1+\beta), 1 < \mu < 2\beta\}, \\
D_2 &= \{(\alpha, \mu) | 2 < \alpha < 2\beta < \mu < \tfrac{1}{4}(\alpha - 2\beta)^2 + 2\beta\}.
\end{aligned}
$$

In the (α, μ) plane of Fig. 1(c), the threshold condition $R_0 = 1$ determines a line

$$C_0 : \mu = 2\beta,$$

which intersects with the boundary of D at $Q(2, 2\beta)$ and $\tilde{Q}(2\beta + 2, 2\beta)$.

The critical condition $\alpha = 2\beta$ in Theorem 3.1 determines another straight line

$$C_1 : \alpha = 2\beta.$$

The intersection of two lines C_0 and C_1 is the point $P(2\beta, 2\beta)$.

Denote the portion of the parabola $\Delta = 0$ by

$$C_2 : \mu = \bar{\mu} := \frac{1}{4}(\alpha - 2\beta)^2 + 2\beta, \quad 2 < \alpha \leqslant 2\beta,$$

which is concave up and tangent to C_0 at the point P. The arc C_2 intersects with boundary lines $\alpha = 2$ of region D at $T(2, \beta^2 + 1)$. Therefore, curves C_0 and C_2 partition the region D into three subregions D_0, D_1 and D_2. If $(\alpha, \mu) \in D_0$, system (2.2) does not have any endemic equilibrium; if $(\alpha, \mu) \in D_1$, system (2.2) has a unique endemic equilibrium E_1; if $(\alpha, \mu) \in D_2$, system (2.2) has two endemic equilibria E_1 and E_2. Equilibria E_1 and E_2 coalesce into one equilibrium on the curve C_2, and disappear when (α, μ) crosses C_2 from D_2 to D_0. Equilibrium E_2 merges with E_0 on the line segment PQ, and E_1 merges with E_0 on the line segment $P\tilde{Q}$. E_1 and E_2 merge with E_0 simultaneously at the point P.

3.2. *Stability of disease-free equilibrium and backward bifurcation*

In this section, we mainly study the stability of disease-free equilibrium E_0 which always exists. In fact, the characteristic equation of system (2.2) at E_0 is

$$(\lambda + 1)\left[\lambda^2 + (\alpha + \mu - 2\beta)\lambda + 2(\mu - 2\beta)\right] = 0.$$

By considering the distribution of the roots of the equation above, one can obtain that the disease-free equilibrium E_0 is locally asymptotically stable (resp. unstable) for $R_0 < 1$ (resp. $R_0 > 1$), i.e., stability of E_0 changes as R_0 passes through the threshold $R_0 = 1$.

To determine the stability of E_0 when $R_0 = 1$ and study the bifurcation near $R_0 = 1$, we choose R_0 as the parameter, and have the following result.

Theorem 3.2. *Consider R_0 as a bifurcation parameter. For $\beta > 1$, system (2.2) undergoes a backward (resp. forward) bifurcation at $R_0 = 1$ if $2 < \alpha < 2\beta$ (resp. $2\beta < \alpha < 2(1 + 2\beta)$).*

Proof. Without loss of generality, we consider R_0 as a function of μ by holding α and β being constants. Let $\mu = 2\beta + \epsilon$, and substitute μ into system (2.2). One should note that $\epsilon = 0$ corresponds to $R_0 = 1$. By the center manifold theorem, system (2.2) reduced on the center manifold reads

$$\dot{X} = \left(-\frac{2}{\alpha}\epsilon + \mathcal{O}(\epsilon^2)\right)X - \left(\frac{4\beta(\alpha - 2\beta)}{\alpha} + \mathcal{O}(\epsilon)\right)X^2 + \mathcal{O}\left(X^3\right). \tag{3.1}$$

Let the right side of equation (3.1) be $\bar{F}(X, \epsilon)$, one can verify that

$$\bar{F}(0,0) = 0, \;\; \bar{F}_X(0,0) = 0, \;\; \bar{F}_\epsilon(0,0) = 0,$$
$$\bar{F}_{X\epsilon}(0,0) = -\frac{2}{\alpha}, \;\; \bar{F}_{XX}(0,0) = -\frac{8\beta(\alpha - 2\beta)}{\alpha}.$$

Hence, system (2.2) undergoes a transcritical bifurcation if $\alpha \neq 2\beta$. Since R_0 decreases in ϵ, when R_0 passes through 1, system (2.2) undergoes a forward bifurcation if $2\beta < \alpha < 2(1+2\beta)$, and undergoes a backward bifurcation if $2 < \alpha < 2\beta$. $\square$

Remark 3.1. Let $\epsilon = 0$ in system (3.1), one can see that E_0 is a saddle-node if $R_0 = 1$ and $\alpha \neq 2\beta$, since equation (3.1) reduces to $\dot{X} = -\frac{4\beta(\alpha-2\beta)}{\alpha}X^2 + \mathcal{O}\left(X^3\right)$. If $\beta > 1$ and $\alpha = 2\beta$, according to the center manifold theorem, there exists a one-dimensional center manifold

$$Y = -\frac{4\beta(\alpha^2 - 2\alpha\beta - \alpha + 4\beta)}{(\alpha-1)\alpha^2}X^2 + \mathcal{O}\left(X^3\right), \quad Z = \frac{8\beta^2(\alpha-1-\beta)}{\alpha-1}X^2 + \mathcal{O}\left(X^3\right),$$

on which equation (3.1) reduces to

$$\dot{X} = -4\beta X^3 + \mathcal{O}\left(X^4\right).$$

Hence, E_0 is a stable semi-hyperbolic node of codimension 2.

If we revisit Fig. 1(c), by Remark 3.1 system (2.2) has E_0 as a semi-hyperbolic node of codimension 2 at the point P, which separates C_0 into two parts. System (2.2) undergoes a forward bifurcation and a backward bifurcation on the line segments $P\tilde{Q}$ and PQ, respectively.

3.3. Stability of endemic equilibria and saddle-node bifurcation

In this subsection we determine the stability of endemic equilibria.

The characteristic equation of system (2.2) at E_i $(i = 1, 2)$ is of the form

$$P_{E_i}(\lambda) := \lambda^3 + a_1(x_i)\lambda^2 + a_2(x_i)\lambda + a_3(x_i) = 0, \tag{3.2}$$

where

$$\begin{aligned} a_1(x_i) &= 4(x_i - 1) + \alpha + 1 > 0, \\ a_2(x_i) &= x_i\left[4(x_i - 1) + \alpha + 2\beta - \mu\right], \\ a_3(x_i) &= 2x_i(x_i - 1)\left[2(x_i - 1) + \alpha - 2\beta\right], \end{aligned} \tag{3.3}$$

in which $x_i > 1$ $(i = 1, 2)$. By Routh–Hurwitz criterion, E_i $(i = 1, 2)$ is stable if and only if

$$a_1(x_i) > 0, \quad a_3(x_i) > 0, \quad \Delta_2(x_i) := a_1(x_i)a_2(x_i) - a_3(x_i) > 0.$$

Theorem 3.3. *For system (2.2) with $(\alpha, \beta, \mu) \in \Lambda$,*

(1) *if $R_0 > 1$, the unique endemic equilibrium E_1 is locally asymptotically stable;*
(2) *if $R_0 = 1$ and $2 < \alpha < 2\beta$, the unique endemic equilibrium E_1 is locally asymptotically stable;*
(3) *if $R_0 < 1$, $2 < \alpha < 2\beta$ and $\Delta > 0$, E_2 is unstable and E_1 is locally asymptotically stable (resp. unstable) if $\Delta_2 > 0$ (resp. $\Delta_2 < 0$).*

Proof. (1) If $R_0 > 1$, then $\Delta > 0$. By algebraic manipulations, we have

$$a_3(x_1) = 2x_1(x_1 - 1)\sqrt{\Delta} > 0,$$
$$\Delta_2(x_1) = x_1[12(x_1-1)^2 + (6\alpha + 4(3\beta - \mu + 1))(x_1 - 1) + (\alpha+1)(\alpha + 2\beta - \mu)] > 0.$$

Therefore, E_1 is locally asymptotically stable.

(2) If $R_0 = 1$ and $2 < \alpha < 2\beta$, then $x_1 = 2\beta - \alpha + 1 > 1$ and

$$\begin{aligned} a_3(x_1) &= 2x_1(2\beta-\alpha)^2 > 0, \\ \Delta_2(x_1) &= x_1\left(4\beta^2 + 2\beta + (26\beta - 7\alpha + 3)(2\beta - \alpha)\right) > 0. \end{aligned}$$

Thus, E_1 is locally asymptotically stable.

(3) If $R_0 < 1$, $2 < \alpha < 2\beta$ and $\Delta > 0$, we can derive that $a_3(x_2) = -2x_2(x_2 - 1)\sqrt{\Delta} < 0$. Since $a_1(x_2) > 0$, the characteristic equation (3.2) at E_2 has one positive root and two roots with negative real part, which implies that E_2 is always unstable.

For equilibrium E_1, by calculations we have $a_3(x_1) = 2x_1(x_1-1)\sqrt{\Delta} > 0$. If $\Delta_2(x_1) > 0$, according to Routh–Hurwitz criterion, E_1 is a stable node or focus; If $\Delta_2(x_1) < 0$, the characteristic equation (3.2) has one negative root and two roots with positive real part, so E_1 is unstable. □

From Theorem 3.1, if $R_0 < 1$, $2 < \alpha < 2\beta$ and $\Delta = 0$ (i.e., $\mu = \bar{\mu}$), two distinct endemic equilibria E_1 and E_2 coalesce at one point, say $E^*(x_c, y_c, z_c)$, where

$$x_c = \frac{2\beta + 2 - \alpha}{2}, \quad y_c = \frac{x_c - 1}{\beta x_c}, \quad z_c = \frac{\bar{\mu}}{2\beta x_c}.$$

From (3.3), the coefficients of characteristic equation at E^* satisfy

$$a_1(x_c) = 4\beta + 1 - \alpha > 0, \;\; a_2(x_c) = x_c(6\beta - \alpha - \bar{\mu}), \;\; a_3(x_c) = 0.$$

In order to show that E^* is a saddle-node below, we first define a subthreshold $R_0^c := R_{0|2<\alpha<2\beta,\Delta=0}$. Noting that $\Delta = (\alpha - 2\beta)^2 - 8\beta(\frac{1}{R_0} - 1)$, a direct calculation leads to

$$R_0^c = \frac{8\beta}{(\alpha - 2\beta)^2 + 8\beta}.$$

Thus $0 < R_0^c < 1$. By Theorem 3.1, system (2.2) has two endemic equilibria if $R_0^c < R_0 < 1$, and has no endemic equilibrium if $R_0 < R_0^c$.

Theorem 3.4. *Consider R_0 as a bifurcation parameter. System (2.2) undergoes a saddle-node bifurcation at $R_0 = R_0^c$ if $2 < \alpha < 2\beta$ and $a_2(x_c) \neq 0$.*

Proof. If $2 < \alpha < 2\beta$ and $R_0 = R_0^c$, two equilibrium points E_1 and E_2 coalesce at $E^*(x_c, y_c, z_c)$. The Jacobian of system (2.2) at E^* has three eigenvalues

$$0,\ \text{and}\ \theta_{1,2} = \frac{-a_1(x_c) \pm \sqrt{a_1^2(x_c) - 4a_2(x_c)}}{2}.$$

The zero eigenvalue is simple if $a_2(x_c) \neq 0$, i.e., $6\beta - \alpha - \bar{\mu} \neq 0$. We linearize system (2.2) at E^* and diagonalize the linear part by an affine map $(x, y, z) \to (u, v, w)$ which depends on the sign of $a_1^2(x_c) - 4a_2(x_c)$. See Appendix A for the three different affine maps when $a_1^2(x_c) - 4a_2(x_c)$ is positive, zero and negative, respectively.

According to the center manifold theory, there exists a one-dimensional center manifold $v = \mathcal{O}(u^2)$, $w = \mathcal{O}(u^2)$ where $u \sim 0$, such that the dynamics on the local center manifold of E^* is governed by the scalar equation

$$\dot{u} = -\frac{\beta(2\beta - \alpha)(2\beta - \alpha + 2)^3}{8a_2(x_c)}u^2 + \mathcal{O}(u^3),$$

which is independent of the sign of $a_1^2(x_c) - 4a_2(x_c)$.

Note that $a_2(x_c) \neq 0$, so the coefficient of u^2 in the equation above is well defined and nonzero. Hence, E^* is a saddle-node point at $R_0 = R_0^c$, and system (2.2) undergoes a saddle-node bifurcation when R_0 passes through the critical value R_0^c. □

If $a_2(x_c) \neq 0$ fails in Theorem 3.4, we will show that E^* is a cusp point in the following section. In all, from the analysis in Sections 3.2 and 3.3, system (2.2) presents different dynamics as R_0 changes.

4. Bifurcations and complex dynamics

4.1. Bogdanov–Takens bifurcation

According to Theorem 3.1, system (2.2) has two distinct endemic equilibria E_1 and E_2 when $\beta > 1$ and $(\alpha, \mu) \in D_2$. On the arc C_2 (i.e., $\Delta = 0$ or $\mu = \bar{\mu}$), E_1 and E_2 coalesce at the point $E^*(x_c, y_c, z_c)$. In Theorem 3.4 we consider the case that $\Delta = 0$ but $a_2(x_c) \neq 0$. If $a_2(x_c) = 0$, the eigenvalues of E^* are $\lambda_{1,2} = 0$ and $\lambda_3 = -a_1(x_c) < 0$. That is to say, when $\Delta = 0$ and $a_2 = 0$, i.e.,

$$(\alpha - 2\beta)^2 + 4(2\beta - \mu) = 0,\ \ 6\beta - \alpha - \frac{1}{4}(\alpha - 2\beta)^2 - 2\beta = 0, \tag{4.1}$$

system (2.2) may exhibit Bogdanov–Takens bifurcation.

We represent α and μ in terms of β by solving system (4.1), which take the forms

$$\begin{aligned} \alpha &= 2(\beta - 1 - \sqrt{2\beta + 1}) := \alpha^*, \\ \mu &= 2(2\beta + 1 + \sqrt{2\beta + 1}) := \mu^*. \end{aligned} \tag{4.2}$$

Since $\alpha^* > 2$, we have $\beta > 3 + \sqrt{6}$. Note that the other root $\alpha = 2(\beta - 1 + \sqrt{2\beta + 1})$ is discarded since it contradicts to the condition $\alpha < 2\beta$.

If $(\alpha, \mu) = (\alpha^*, \mu^*)$, we have

$$E^*: \ (x_c, y_c, z_c) = \left(2 + \sqrt{2\beta + 1}, \ \frac{2}{x_c(x_c - 3)}, \ \frac{2(x_c - 2)}{x_c(x_c - 3)}\right),$$

where $x_c > 3 + \sqrt{6}$. The Jacobian matrix of system (2.2) at E^* is

$$J(E^*) = \begin{bmatrix} -1 & \frac{x_c^2(x_c-1)(x_c-3)}{2} & 0 \\ \frac{2(x_c-1)(2x_c-5)}{x_c^2(x_c-3)} & -(x_c - 1)(x_c - 3) & 2(x_c - 2) \\ -\frac{4(x_c-1)(x_c-2)}{x_c^2(x_c-3)} & 0 & -2(x_c - 1) \end{bmatrix}.$$

In this case, the characteristic equation of system (2.2) at E^* has 3 roots

$$\lambda_{1,2} = 0, \ \lambda_3 = -\left(x_c^2 - 2x_c + 2\right) = -\left(2\beta + 3 + 2\sqrt{2\beta + 1}\right) < 0.$$

Theorem 4.1. *For $\beta > 3 + \sqrt{6}$ and $(\alpha, \mu) = (\alpha^*, \mu^*)$, the unique endemic equilibrium $E^*(x_c, y_c, z_c)$ is a cusp point of codimension 2, and system (2.2) near E^* is topologically equivalent to*

$$\begin{cases} \dot{u} = v, \\ \dot{v} = u^2 + uv + \mathcal{O}\left(|u, v|^3\right). \end{cases} \tag{4.3}$$

Proof. Substituting $\alpha = \alpha^*$ and $\mu = \mu^*$ into system (2.2) yields

$$\begin{cases} \dot{x} = x(1 - x + \beta xy), \\ \dot{y} = 2(1 - z) - 6\beta y + \beta xy(2z - y), \\ \dot{z} = 2y\left(2\beta + 1 + \sqrt{2\beta + 1} - \beta xz\right). \end{cases}$$

Now introduce an affine transformation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} + \begin{bmatrix} -\frac{x_c^2(x_c-3)}{2(x_c-2)} & \frac{x_c^2(x_c-3)}{2(x_c-2)} & \frac{x_c^2(x_c-2)(x_c-3)}{4(x_c-1)} \\ \frac{-1}{(x_c-1)(x_c-2)} & 0 & \frac{-(x_c-2)}{2} \\ 1 & -\frac{2x_c-1}{2(x_c-1)} & 1 \end{bmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}, \tag{4.4}$$

and we obtain the following system

$$\begin{cases} \dot{u} = v + L_{20}u^2 + L_{11}uv + L_{02}v^2 + \mathcal{O}\left(|u, v, w|^3, w|u, v, w|\right), \\ \dot{v} = M_{20}u^2 + M_{11}uv + M_{02}v^2 + \mathcal{O}\left(|u, v, w|^3, w|u, v, w|\right), \\ \dot{w} = \lambda_3 w + \mathcal{O}\left(|u, v, w|^2\right), \end{cases} \tag{4.5}$$

whose linear part is in Jordan canonical form of $J(E^*)$, and

$$L_{20} = \frac{x_c\,(x_c-3)\left(4\,x_c^5-19\,x_c^4+42\,x_c^3-54\,x_c^2+40\,x_c-12\right)}{2\lambda_3^2(x_c-2)}, \qquad M_{20} = -\frac{x_c^3(x_c-1)(x_c-3)}{\lambda_3(x_c-2)},$$
$$L_{11} = -\frac{x_c\,(x_c-3)\left(12\,x_c^5-65\,x_c^4+146\,x_c^3-174\,x_c^2+112\,x_c-32\right)}{2\lambda_3^2(x_c-2)}, \quad M_{11} = \frac{x_c\,(x_c-3)\left(4\,x_c^3-13\,x_c^2+14\,x_c-6\right)}{\lambda_3(x_c-2)},$$
$$L_{02} = \frac{x_c\,(x_c-1)(x_c-3)(2\,x_c-1)\left(4\,x_c^2-9\,x_c+6\right)}{2\lambda_3^2}, \qquad M_{02} = -\frac{x_c\,(x_c-1)(x_c-3)\left(3\,x_c^2-9\,x_c+5\right)}{\lambda_3(x_c-2)}.$$

By center manifold theorem, system (4.5) restricted to the center manifold $W^c_{loc}(E^*)$ becomes

$$\begin{cases} \dot{u} = v + L_{20}u^2 + L_{11}uv + L_{02}v^2 + \mathcal{O}\left(|u,v|^3\right), \\ \dot{v} = M_{20}u^2 + M_{11}uv + M_{02}v^2 + \mathcal{O}\left(|u,v|^3\right). \end{cases} \tag{4.6}$$

Using a near-identity transformation $u = X + \cdots$ and $v = Y + \cdots$, we can eliminate non-resonant terms of system (4.6). We rewrite X, Y into u, v, and obtain

$$\begin{cases} \dot{u} = v, \\ \dot{v} = M_{20}u^2 + (M_{11} + 2L_{20})uv + \mathcal{O}\left(|u,v|^3\right). \end{cases} \tag{4.7}$$

Next we consider the coefficients M_{20} and M_{11} in the second equation of system (4.7). Since $x_c > 3 + \sqrt{6}$,

$$M_{20} = -\frac{x_c^3\,(x_c-1)\,(x_c-3)}{\lambda_3\,(x_c-2)} > 0,$$

and

$$M_{11} + 2L_{20} = \frac{2x_c^3\,(x_c-3)\left(x_c^2-3x_c+3\right)}{\lambda_3^2\,(x_c-2)} > 0.$$

We rescale the variables and time

$$(u,v,t) \to \left(\frac{M_{20}}{(M_{11}+2L_{20})^2}u,\ \frac{M_{20}^2}{(M_{11}+2L_{20})^3}v,\ \frac{(M_{11}+2L_{20})}{M_{20}}t\right),$$

then system (4.7) is topologically equivalent to the normal form (4.3), and E^* is a cusp point of codimension 2. $\square$

From Theorem 4.1, we need two parameters to unfold the codimension 2 singularity E^*. In the following we consider (α, μ) as bifurcation parameters and perturb system (2.2) with (α, μ) in a small neighborhood of (α^*, μ^*). Substituting

$$\alpha = \alpha^* + \varepsilon_1, \quad \mu = \mu^* + \varepsilon_2$$

into system (2.2), we obtain a perturbed system

$$\begin{cases} \dot{x} = x\left(1 - x + \beta xy\right), \\ \dot{y} = 2(1 - z) - \left(6\beta + \varepsilon_1 + \varepsilon_2\right) y + \beta xy(2z - y), \\ \dot{z} = y\left(\mu^* + \varepsilon_2 - 2\beta xz\right), \end{cases} \tag{4.8}$$

for parameters $(\varepsilon_1, \varepsilon_2)$ in a small neighborhood of $(0, 0)$.

Theorem 4.2. *For $\beta > 3 + \sqrt{6}$ and parameters $\varepsilon = (\varepsilon_1, \varepsilon_2)$ sufficiently small, system (4.8) is a generic unfolding of the cusp singularity of codimension 2. The singularity of system (4.8) has a universal unfolding*

$$\begin{cases} \dot{u} = v, \\ \dot{v} = \nu_1(\varepsilon) + \nu_2(\varepsilon)\, v + u^2 + uv + v\mathcal{O}\left(|u, v|^2\right), \end{cases} \tag{4.9}$$

where $\nu_1(\varepsilon)$ and $\nu_2(\varepsilon)$ have the form

$$\begin{aligned} \nu_1(\varepsilon) &= \tfrac{16\, x_c^2\, \left(x_c^2 - 3\, x_c + 3\right)^4}{\lambda_3^6 (x_c - 1)^3} \left[4\left(x_c - 1\right)^2 \varepsilon_1 + 4\left(x_c - 1\right)\varepsilon_2 + \varepsilon_1\varepsilon_2 + \mathcal{O}(\varepsilon^3)\right], \\ \nu_2(\varepsilon) &= \tfrac{2\left(x_c^2 - 3\, x_c + 3\right)}{\lambda_3^3 (x_c - 1)^2} \left[\left(x_c - 1\right)\left(16\, x_c^5 - 102\, x_c^4 + 239\, x_c^3 - 256\, x_c^2 + 124\, x_c - 24\right)\varepsilon_1 \right. \\ &\quad \left. + \left(16\, x_c^5 - 103\, x_c^4 + 243\, x_c^3 - 262\, x_c^2 + 128\, x_c - 24\right)\varepsilon_2 + \mathcal{O}(\varepsilon^2)\right], \end{aligned} \tag{4.10}$$

with

$$\left| \frac{\partial\left(\nu_1,\ \nu_2\right)}{\partial\left(\varepsilon_1,\ \varepsilon_2\right)} \right|_{\varepsilon=0} = \frac{128\sqrt{2\beta + 1}(2 + \sqrt{2\beta + 1})^3(2 + 2\beta + \sqrt{2\beta + 1})^5}{(1 + \sqrt{2\beta + 1})^3(3 + 2\beta + 2\sqrt{2\beta + 1})^8} \neq 0.$$

Proof. See Appendix B. □

By Theorem 4.2, system (4.8) is a generic family unfolding the cusp singularity of codimension 2, and Bogdanov–Takens bifurcation of codimension 2 occurs at $(\varepsilon_1, \varepsilon_2) = (0, 0)$. Immediately, we know that Hopf bifurcation of codimension 1 and homocline bifurcation of codimension 1 occur in a small neighborhood of $(\varepsilon_1, \varepsilon_2) = (0, 0)$, or equivalently, in a small neighborhood of (α^*, μ^*) in the (α, μ) plane [4,29]. In a small open cone with vertex (α^*, μ^*) bounded by the curve of Hopf bifurcation and the curve of homoclinic bifurcation, system (2.2) has a unique limit cycle, which is unstable. In the following, we will trace this unstable limit cycle by numerical simulations.

Fix parameter $\beta = 11.711918$. According to the formula (4.2),

$$(\alpha^*, \mu^*) \approx (11.53974049, 58.73176751)$$

is the unique Bogdanov–Takens point of codimension 2 in (α, μ) plane. Furthermore, we freeze $\alpha = 11$ close to α^*, change μ continuously, and plot the bifurcation diagram in Fig. 2(a), from which one can see that the unstable limit cycle exists on the interval $\mu \in (61.266, 61.540)$, which appears via a homoclinic bifurcation at $\mu = 61.266$ and terminates via a subcritical Hopf bifur-

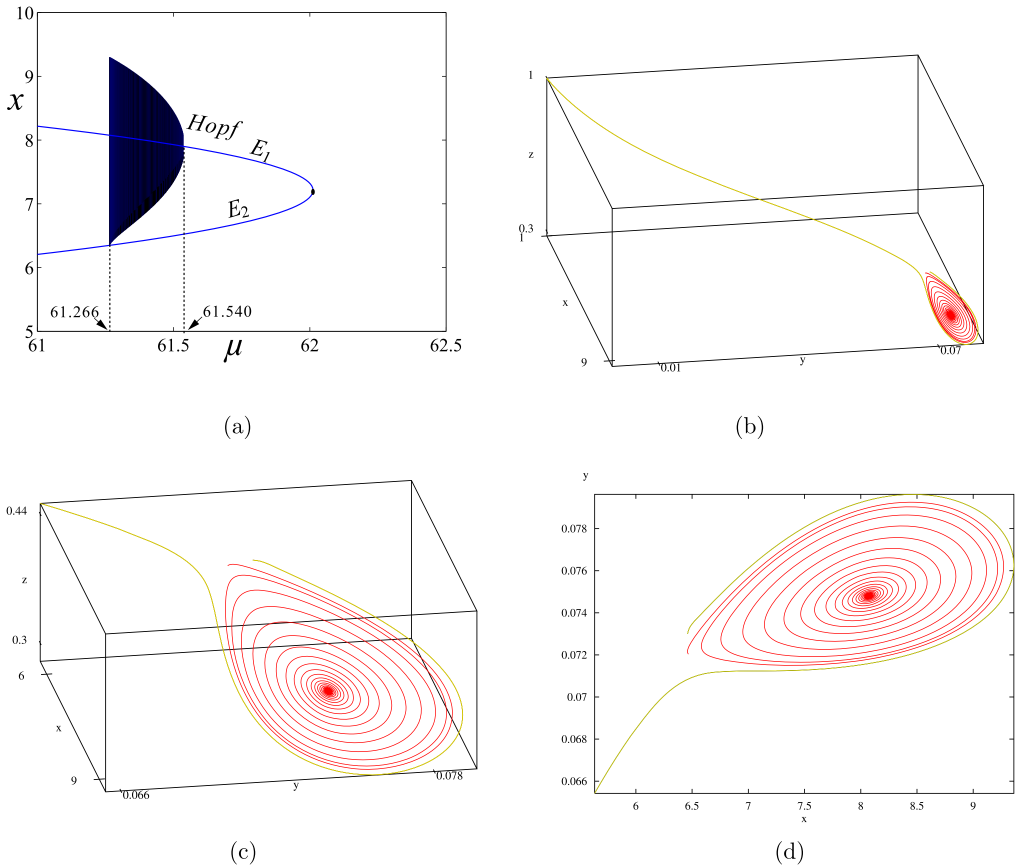


Fig. 2. (a) Bifurcation diagram in (μ, x) plane with $\beta = 11.711918$ and $\alpha = 11$. The unstable limit cycle appears via a homoclinic bifurcation at $\mu \approx 61.266$ and terminates via a subcritical Hopf bifurcation at $\mu \approx 61.540$. The endemic equilibrium E_1 lose stability for $\mu \geqslant 61.540$ through the subcritical Hopf bifurcation. E_2 is always a hyperbolic saddle. (b) Phase portrait of system (2.2) with $\beta = 11.711918$, $\alpha = 11$ and $\mu = 61.27$. (c) A magnified view of the bottom-right corner of sub-figure (b). (d) Projection of solution curves in sub-figure (c) onto xy plane. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

cation at $\mu = 61.540$. The amplitude of the limit cycle decreases as μ increases. The upper and lower branches of curve opening left represent the endemic equilibria E_1 and E_2, respectively.

From Fig. 2 (a), we know that for any given $\mu \in (61.266, 61.540)$, system (2.2) has an unstable limit cycle surrounding the stable endemic equilibrium E_1. As an example, we choose $\mu = 61.27$, an interior point of the interval $(61.266, 61.540)$, and sketch the phase portrait. In this case, system (2.2) has a stable disease-free equilibrium $E_0(1, 0, 1)$, a stable endemic equilibrium $E_1(8.0732, 0.0748, 0.3240)$, and an unstable endemic equilibrium $E_2(6.3507, 0.0719, 0.4119)$. We choose two sets of initial conditions and plot solution curves. The orbit with initial condition $(6.46691, 0.07203577, 0.4039)$ converges to E_1 (red curve), and the orbit with initial condition $(6.46, 0.073, 0.408)$ converges to E_0 (green curve), which implies that there is an unstable limit cycle lying in a neighborhood of E_1. See Fig. 2(b)–(d).

4.2. Hopf bifurcations and multiple limit cycles

4.2.1. Transversality condition

In this section, we will investigate periodic solutions generated from Hopf bifurcation. By the stability analysis in Theorem 3.3 and Figs. 1(a) and 1(b), we can see that Hopf bifurcation will not occur for $\beta \leqslant 1$. For $\beta > 1$, if $(\alpha, \mu) \in D_0$ or D_1, any equilibrium point, whenever exists, will never lose stability, so Hopf bifurcation could only occur in region D_2. Hence, we assume that

$$\beta > 1, \ (\alpha, \mu) \in D_2$$

throughout this section. Under this condition, system (2.2) has two endemic equilibria E_1 and E_2. By Theorem 3.3, E_2 is always a hyperbolic saddle and Hopf bifurcation cannot occur at E_2. In summary, Hopf bifurcation can occur only when the stability of E_1 changes. Since the characteristic polynomial $P_{E_1}(\lambda)$ has a pair of pure imaginary roots if and only if $\Delta_2(x)_{|x=x_1} = 0$, we first study the function $\Delta_2(x) = 0$. Denote

$$M(x) = 12(x-1)^2 + (6\alpha + 12\beta - 4\mu + 4)\,(x-1) + (\alpha+1)(\alpha + 2\beta - \mu),$$

and then $\Delta_2(x) = xM(x)$. Define

$$\xi := 6\beta - \alpha - \mu,$$

and

$$x_H := \frac{-(6\alpha + 12\beta - 4\mu + 4) + \sqrt{\Delta_1}}{24} + 1,$$

where

$$\Delta_1 = (6\alpha + 12\beta - 4\mu + 4)^2 - 48(\alpha+1)(\alpha + 2\beta - \mu).$$

Theorem 4.3. *For system* (2.2) *with $\beta > 1$ and $(\alpha, \mu) \in D_2$,*

(1) *if $\xi \geqslant 0$, E_1 is locally asymptotically stable;*
(2) *if $\xi < 0$, a generic Hopf bifurcation could occur at E_1 when $x_1 = x_H$.*

Proof. (1) Substituting x_1 into $M(x)$, we obtain

$$\Delta_2(x_1) = x_1[(4\beta + 1 - \alpha)\xi + 3\Delta + (3\xi + \mu + 2)\sqrt{\Delta}],$$

Note that $(\alpha, \mu) \in D_2$, so $2\beta > \alpha$. Hence, $\Delta_2(x_1) > 0$ if $\xi \geqslant 0$. By part (3) in Theorem 3.3, E_1 is locally asymptotically stable.

(2) For $\beta > 1$ and $(\alpha, \mu) \in D_2$, we can see that $\xi < 0$ implies

$$\alpha + 2\beta < 6\beta - \alpha < \mu.$$

Hence, $\alpha + 2\beta - \mu < 0$ and $x_H > 1$.

If $x_1 = x_H$, then $\Delta_2(x_1) = 0$ and $P_{E_1}(\lambda) = 0$ has a pair of pure imaginary roots. For the transversality condition, we consider μ as the bifurcation parameter and suppose that there exists μ_H such that $\Delta_2(x_1(\mu_H), \mu_H) = 0$. Then $M(x_1(\mu_H), \mu_H) = 0$ and

$$\begin{aligned}\frac{d\Delta_2(x_1(\mu),\mu)}{d\mu}\Big|_{\mu=\mu_H} &= x_1(\mu_H)\left[\frac{\partial M(x_1(\mu),\mu)}{\partial x_1}\frac{\partial x_1(\mu)}{\partial \mu} + \frac{\partial M(x_1(\mu),\mu)}{\partial \mu}\right]\Big|_{\mu=\mu_H} \\ &= x_H\left[-\frac{\sqrt{\Delta_1(\mu_H)}}{\sqrt{\Delta(\mu_H)}} - \Big(4(x_H-1)+\alpha+1\Big)\right] < 0.\end{aligned}$$

The proof is complete. □

Remark 4.1. If $\beta > 1$, $(\alpha, \mu) \in D_2$ and $\xi < 0$, then $6\beta - \alpha < \mu < \frac{1}{4}(\alpha - 2\beta)^2 + 2\beta$ from which we can derive the relations between α and β:

$$\alpha < 2(\beta - 1 - \sqrt{2\beta + 1}).$$

Recall that $\alpha > 2$, thus the inequality above implies $\beta > 3 + \sqrt{6}$. Note that the other inequality $\alpha > 2(\beta - 1 + \sqrt{2\beta + 1})$ is discarded since it contradicts $\alpha < 2\beta$. In all, Hopf bifurcation occurs only if

$$2 < \alpha < 2(\beta - 1 - \sqrt{2\beta + 1}) < \alpha + 2\beta < 6\beta - \alpha < \mu < \frac{1}{4}(\alpha - 2\beta)^2 + 2\beta.$$

4.2.2. Degenerate Hopf bifurcations

By part (2) of Theorem 4.3, Hopf bifurcation could occur when $\Delta_2(x_1) = 0$ (i.e., $x_1 = x_H$). To study the number and stability of limit cycles generated from Hopf bifurcation, we need to calculate the first Lyapunov coefficient l_1 by deriving the normal form of Hopf bifurcation, which can be achieved by the following procedure. Applying the affine map

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \begin{bmatrix} 1 & 0 & 1 \\ \frac{1}{\beta x_1^2} & -\frac{\sqrt{a_2(x_1)}}{\beta x_1^2} & -\frac{4(x_1-1)+\alpha}{\beta x_1^2} \\ -\frac{(x_1-1)(2x_1+\alpha-1)}{\beta x_1^2(x_1-2)} & -\frac{\sqrt{a_2(x_1)}(2x_1+\alpha-1)}{2\beta x_1^2(x_1-2)} & -\frac{a_2(x_1)+4(x_1-1)^2}{2\beta x_1^2(x_1-2)} \end{bmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix},$$

we obtain

$$\begin{cases} \dot{u} = -\sqrt{a_2(x_1)}v + \sum\limits_{i+j+k\geqslant 2} a_{ijk}u^i v^j w^k, \\ \dot{v} = \sqrt{a_2(x_1)}u + \sum\limits_{i+j+k\geqslant 2} b_{ijk}u^i v^j w^k, \\ \dot{w} = -a_1(x_1)w + \sum\limits_{i+j+k\geqslant 2} c_{ijk}u^i v^j w^k, \end{cases} \tag{4.11}$$

where a_{ijk}, b_{ijk} and c_{ijk} are coefficients depending on α, β and μ. Here we do not present the long expressions of coefficients. The local center manifold of E_1 takes the form

$$w = h_{20}u^2 + h_{11}uv + h_{02}v^2 + \mathcal{O}(|u, v|^3),$$

where

$$\begin{aligned} h_{20} &= \frac{(a_1^2 + 2a_2)c_{200} + 2a_2c_{020} - a_1\sqrt{a_2}c_{110}}{a_1(a_1^2 + 4a_2)}, \\ h_{11} &= \frac{2\sqrt{a_2}(c_{200} - c_{020}) + a_1c_{110}}{a_1^2 + 4a_2}, \\ h_{02} &= \frac{2a_2c_{200} + (a_1^2 + 2a_2)c_{020} + a_1\sqrt{a_2}c_{110}}{a_1(a_1^2 + 4a_2)}. \end{aligned}$$

System (4.11) restricted on the center manifold is

$$\begin{cases} \dot{u} = -\sqrt{a_2(x_1)}v + \sum\limits_{i+j=2} a_{ij0}u^iv^j + \sum\limits_{i+j=3} d_{ij}u^iv^j + \mathcal{O}\left(|u, v|^4\right), \\ \dot{v} = \sqrt{a_2(x_1)}u + \sum\limits_{i+j=2} b_{ij0}u^iv^j + \sum\limits_{i+j=3} e_{ij}u^iv^j + \mathcal{O}\left(|u, v|^4\right), \end{cases}$$

where

$$\begin{aligned} d_{12} &= h_{11}a_{011} + h_{02}a_{101} + a_{120}, & d_{21} &= h_{20}a_{011} + h_{11}a_{101} + a_{210}, \\ e_{12} &= h_{11}b_{011} + h_{02}b_{101} + b_{120}, & e_{21} &= h_{20}b_{011} + h_{11}b_{101} + b_{210}, \\ d_{30} &= h_{20}a_{101} + a_{300}, & d_{03} &= h_{02}a_{011} + a_{030}, \\ e_{30} &= h_{20}b_{101} + b_{300}, & e_{03} &= h_{02}b_{011} + b_{030}. \end{aligned}$$

Then the first Lyapunov coefficient l_1 has the representation [12]

$$\begin{aligned} l_1 =& \frac{1}{8}(a_{120} + b_{210} + 3a_{300} + 3b_{030} + 3h_{20}a_{101} + h_{02}a_{101} + h_{11}a_{011} + h_{20}b_{011} \\ &+ 3h_{02}b_{011} + h_{11}b_{101}) \\ &+ \frac{1}{8\sqrt{a_2(x_1)}}\left(a_{110}a_{020} + a_{110}a_{200} + 2a_{020}b_{020} - 2a_{200}b_{200} - b_{110}b_{020} - b_{110}b_{200}\right). \end{aligned}$$

One should notice that all coefficients in the formula above are functions of α, β and μ. Due to complexity, we choose not to present l_1 in terms of α, β and μ here, because it is difficult to write l_1 in a concise form to determine its sign. At first glance, the complicated expression of l_1 seems not so helpful to obtain information regrading the Hopf bifurcation. However, it turns out that, with the help of simulation, it is quite useful for us to explore oscillating dynamics of system (2.2) and the region of multiple limit cycles by tracing the sign of l_1.

Fix $\beta = 11.711918$, as mentioned in Section 4.1, $(\alpha^*, \mu^*) \approx (11.53974049, 58.73176751)$ is the unique point in the (α, μ) plane at which system (2.2) has a cusp singularity of codimension 2. The condition $\Delta_2(\alpha, \mu) := \Delta_2(\alpha, \mu, x_1(\alpha, \mu)) = 0$ defines a curve of Hopf bifurcation C_h in the (α, μ) plane implicitly. It is obvious that $\Delta_2(\alpha^*, \mu^*) = 0$. We sketch such curve in Fig. 3(a), which starts from the point (α^*, μ^*) and ends at the boundary point $(2, 89.488)$ of D_2. In addition, from Fig. 3(a) we see that $\Delta_2(\alpha, \mu) = 0$ defines an implicit function $\mu = \mu(\alpha)$,

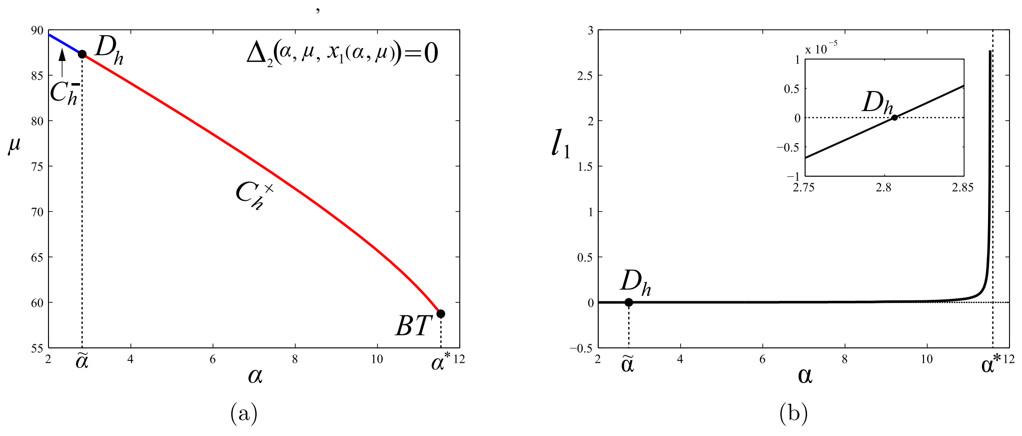


Fig. 3. (a). Curve of Hopf bifurcation C_h in the (α, μ) plane. Supercritical Hopf bifurcation occurs when across the arc C_h^-, and subcritical Hopf bifurcation occurs when across the arc C_h^+. (b). Graph of first Lyapunov coefficient $l_1 = l_1(\alpha)$ for $\alpha \in (2, \alpha^*)$. Here $\beta = 11.711918$.

$\alpha \in (2, \alpha^*)$, s.t. $\Delta_2(\alpha, \mu(\alpha)) = 0$. Substitute $\mu = \mu(\alpha)$ into the first Lyapunov coefficient l_1, then $l_1 = l_1(\alpha) := l_1(\alpha, \mu(\alpha))$. We track the value of $l_1(\alpha)$ as α varies throughout the interval $(2, \alpha^*)$ by plotting the graph of $l_1 = l_1(\alpha)$ in Fig. 3(b). We can observe that $l_1(\alpha)$ changes its sign only once at $\tilde{\alpha} \approx 2.8064294$, and from negative to positive as α increasing. The degenerate Hopf bifurcation may occur at the point

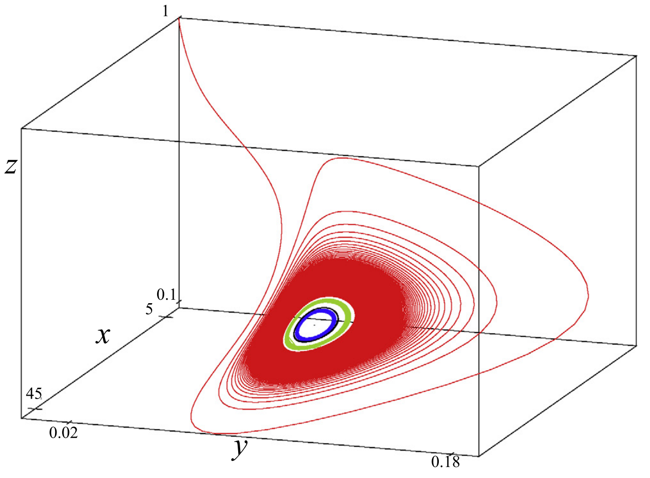
$$D_h : (\tilde{\alpha}, \tilde{\mu}) \approx (2.8064294, 87.3379096)$$

on the curve of Hopf bifurcation C_h. The point $(\tilde{\alpha}, \tilde{\mu})$ divides the curve C_h into two subarcs C_h^- and C_h^+, on which $l_1 < 0$ and $l_1 > 0$, respectively. Supercritical Hopf bifurcation occurs when across the arc C_h^-, and subcritical Hopf bifurcation occurs when across the arc C_h^+.

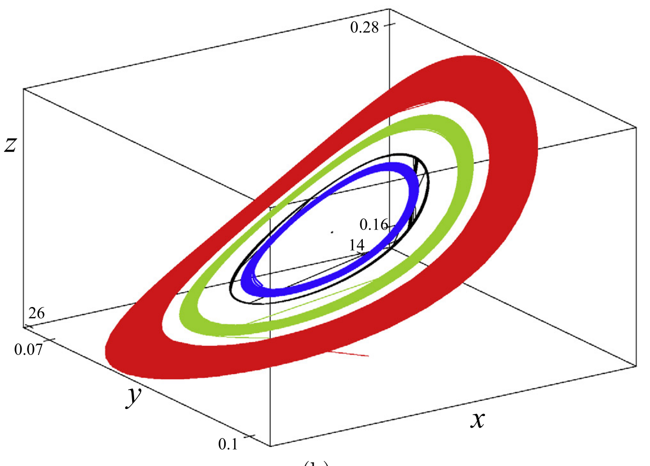
Since $l_1(\alpha)$ vanishes at $\alpha = \tilde{\alpha}$, degenerate Hopf bifurcation may occur at $(\tilde{\alpha}, \tilde{\mu})$ and system (2.2) may exhibits multiple limit cycles for parameters (α, μ) in a small neighborhood of $(\tilde{\alpha}, \tilde{\mu})$. In the following simulation, we show the existence of two limit cycles near $(\tilde{\alpha}, \tilde{\mu})$.

Choose a point $(\hat{\alpha}, \hat{\mu}) = (2.1, 89.225339)$ close to $(\tilde{\alpha}, \tilde{\mu})$. In this case, E_0 is a stable node and $E_1(18.581, 0.0808, 0.205)$ is an unstable focus. The orbit (blue curve) with initial condition $(20.9, 0.08, 0.18)$ spirals outward to the stable limit cycle (black curve). The orbit (green curve) with initial condition $(22.89417, 0.08933377, 0.1738676)$ spirals inward to the stable limit cycle. The orbit (red curve) initially at $(21.9, 0.1, 0.18)$ spirals outward and converges to the stable equilibrium E_0. Since limit cycles lie on the two dimensional local center manifold of E_1, in this example there are two limit cycles, the inner limit cycle is stable and the outer one is unstable. See Fig. 4.

In Section 4.1, we have proved that the Bogdanov–Takens bifurcation of codimension 2 occurs when $(\alpha, \mu) = (\alpha^*, \mu^*)$ which guarantees the existence of subcritical Hopf bifurcation in a small neighborhood of (α^*, μ^*), by which only one unstable limit cycle can be produced. However, we can not make any conclusion from Bogdanov–Takens bifurcation if the parameters (α, μ) are far from the point (α^*, μ^*). In this subsection, we have shown that the first Lyapunov coefficient l_1 can change its sign and vanishes at the D_h point, for which the order of Hopf bifurcation is



(a)



(b)

Fig. 4. Two limit cycles in a neighborhood of unstable equilibrium of E_1. The inner limit cycle is stable and the outer limit is unstable. Subfigure (b) is a magnified view of (a) in a small neighborhood of E_1. Here $\beta = 11.711918$, $\alpha = 2.1$ and $\mu = 89.225339$.

at least 2, i.e., system can present at least 2 limit cycle in a small neighborhood of D_h point. Therefore, we can conclude that:

Bogdanov–Takens bifurcation point (α^*, μ^*) *is not an organizing center of system* (2.2)*, and Bogdanov–Takens bifurcation of cod-2 and Hopf bifurcation of at least cod-2 are two different mechanisms to generate oscillating dynamics in adaptive-network epidemic models.*

4.3. Bifurcation diagram and simulations

In this section, we consider α and μ as parameters and present bifurcation diagrams in (α, μ) plane. With three different levels of β: (a) $0 < \beta \leqslant \frac{1}{2}$; (b) $\frac{1}{2} < \beta \leqslant 1$; (c) $\beta > 1$, we discussed the existence of equilibria in Section 3.1. According to Theorem 3.3, the disease-free equilibrium E_0 is stable (resp. unstable) if $R_0 < 1$ (resp. $R_0 > 1$), and the unique endemic equilibrium E_1 is

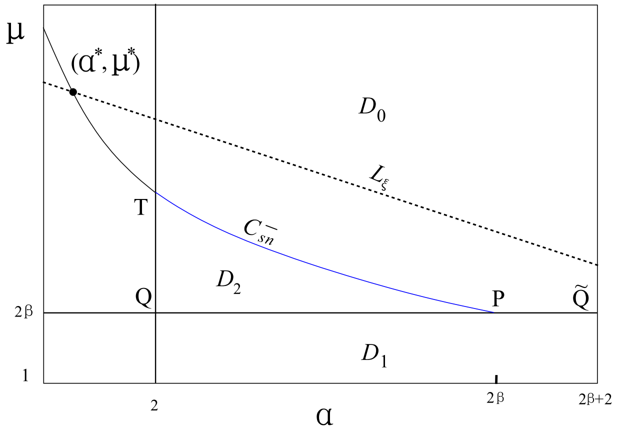


Fig. 5. Bifurcation diagrams for $1 < \beta \leqslant 3+\sqrt{6}$. Bistability phenomenon is present for $(\alpha, \mu) \in D_2$.

stable when $R_0 > 1$. So local bifurcation diagrams of cases (a) and (b) are Figs. 1(a) and 1(b), respectively.

However, for the case $\beta > 1$, system (2.2) may present bistability and oscillations. The dynamics are complicated particularly when $(\alpha, \mu) \in D_2$. There are two distinct endemic equilibria E_1 and E_2 in this case, and the equilibrium E_2 with lower epidemicity is a hyperbolic saddle. By Theorem 4.2 and Remark 4.1, $\beta = 3+\sqrt{6}$ serves as a threshold for the occurrence of Bogdanov–Takens bifurcation and Hopf bifurcation. Therefore, it is necessary to discuss the bifurcation diagrams for $1 < \beta \leqslant 3+\sqrt{6}$ and $\beta > 3+\sqrt{6}$ separately.

Define a straight line L_ξ in the (α, μ) plane,

$$L_\xi : \mu = 6\beta - \alpha.$$

and the domains above and below L_ξ correspond to $\xi < 0$ and $\xi > 0$, respectively.

If $1 < \beta \leqslant 3+\sqrt{6}$, we have $\xi > 0$ for any $(\alpha, \mu) \in D_1 \cup D_2$. According to Theorem 3.3 part (1) and Theorem 4.3 part (1), E_1 is locally asymptotically stable. In fact, L_ξ intersects with the vertical line $\alpha = 2\beta + 2$ at $(2\beta + 2, 4\beta - 2)$. L_ξ and left half part of the parabola $\mu = \frac{1}{4}(\alpha - 2\beta)^2 + 2\beta$ intersect only once at the point $BT(\alpha^*, \mu^*)$ with $\alpha^* \leqslant 2$. Therefore, straight line L_ξ always lies above the region $D_1 \cup D_2$ in the (α, μ) plane. See Fig. 5. By Theorem 4.2 and Remark 4.1, there are no Bogdanov–Takens bifurcation or Hopf bifurcation occurring for $1 < \beta \leqslant 3+\sqrt{6}$. In this case, the bistability phenomenon, i.e., coexistence of stable disease-free equilibrium E_0 and the stable endemic equilibria E_1, is present for $(\alpha, \mu) \in D_2$. The equilibrium E^* is an attracting saddle-node on the arc C_2, so denote C_2 as C_{sn}^-.

If $\beta > 3+\sqrt{6}$, by Theorem 4.2, Bogdanov–Takens bifurcation of codimension 2 occurs for $(\alpha, \mu) = (\alpha^*, \mu^*)$, it has been proven that (α, μ) can be used to unfold the Bogdanov–Takens bifurcation. The bifurcation sets of system (4.9) using ν_1 and ν_2 as unfolding parameters are described in [12]. By Theorem 4.2, system (4.8) has the same bifurcation sets with respect to $(\varepsilon_1, \varepsilon_2)$ as system (4.9) has with respect to (ν_1, ν_2), up to a homeomorphism in the parameter space.

Near a small neighborhood of $(\varepsilon_1, \varepsilon_2) = (0, 0)$, curve of saddle-node bifurcation of system (4.8) is

$$C_{sn}:\ \{(\nu_1(\varepsilon),\nu_2(\varepsilon))\,|\,\nu_1(\varepsilon)=0\}=\{(\varepsilon_1,\varepsilon_2)\,|\,\varepsilon_2=(1-x_c)\,\varepsilon_1+\frac{1}{4}\varepsilon_1^2+\mathcal{O}\left(\varepsilon_1^2\right)\},$$

where $x_c=2+\sqrt{2\beta+1}$. Along the curve C_{sn}, system (4.8) has a unique endemic equilibrium with a simple zero eigenvalue. The Bogdanov–Takens point $BT(\alpha^*,\mu^*)$ separates the saddle-node bifurcation curve into two branches C_{sn}^- and C_{sn}^+ corresponding to $\alpha^*<\alpha<2\beta$ and $2<\alpha<\alpha^*$, respectively. The unstable equilibrium E_1 and the saddle point E_2 coalesce on the curve C_{sn}^+, while the stable equilibrium E_1 and the saddle point E_2 coalesce on the curve C_{sn}^-.

In a small neighborhood of $(\varepsilon_1,\varepsilon_2)=(0,0)$, the endemic equilibrium E_1 undergoes a subcritical Hopf bifurcation on the curve

$$\begin{aligned}C_h^{loc}:\ &\{(\nu_1(\varepsilon),\nu_2(\varepsilon))\,|\,\nu_1(\varepsilon)=-\nu_2^2(\varepsilon),\ \nu_2(\varepsilon)>0\}\\ &=\{(\varepsilon_1,\varepsilon_2)\,|\,\varepsilon_2=(1-x_c)\,\varepsilon_1-\frac{(x_c^3-6\,x_c^2+12\,x_c-10)(x_c^3-2\,x_c^2+2)}{16(x_c^2-3\,x_c+3)^2}\varepsilon_1^2+\mathcal{O}\left(\varepsilon_1^3\right),\varepsilon_1<0\}.\end{aligned}$$

The Hopf bifurcation gives rise to an unstable limit cycle in a small neighborhood of the stable equilibrium E_1. See Fig. 2(a)–(d).

System (4.8) has a homoclinic bifurcation occurring on the curve C_{hom} defined by

$$\begin{aligned}C_{hom}:\ &\{(\nu_1(\varepsilon),\nu_2(\varepsilon))\,|\,\nu_1(\varepsilon)=-\tfrac{49}{25}\nu_2^2(\varepsilon)+\mathcal{O}(\nu_2^{\frac{5}{2}}(\varepsilon)),\ \nu_2(\varepsilon)>0\}\\ &=\{(\varepsilon_1,\varepsilon_2)\,|\,\varepsilon_2=(1-x_c)\,\varepsilon_1-\frac{(7x_c^3-18\,x_c^2+12\,x_c+2)(7x_c^3-38\,x_c^2+72\,x_c-58)}{400(x_c^2-3\,x_c+3)^2}\varepsilon_1^2+\mathcal{O}\left(\varepsilon_1^3\right),\varepsilon_1<0\}.\end{aligned}$$

The limit cycle generated from the subcritical bifurcation disappears from the homoclinic bifurcation.

Remark 4.2. Compared to the analysis in Sections 3.3 and 4.2 for the global saddle-node bifurcation and Hopf bifurcation, it is necessary to note that:

(1) By a direct calculation of substituting $\alpha=\alpha^*+\varepsilon_1$ and $\mu=\mu^*+\varepsilon_2$ into $R_0=R_0^c$ (i.e. $\Delta=0$), one can obtain the curve of the global saddle-node bifurcation $\varepsilon_2=(1-x_c)\,\varepsilon_1+\frac{1}{4}\,\varepsilon_1^2$, which is exactly equal to the Taylor expansion of the curve C_{sn} up to the second order terms. This illustrates that the analysis of saddle-node bifurcation in Section 3.3 is consistent with the result derived from the Bogdanov–Takens bifurcation.

(2) For the Hopf bifurcation, if we substitute $\alpha=\alpha^*+\varepsilon_1$ and $\mu=\mu^*+\varepsilon_2$ into the condition of Hopf bifurcation, resultant$(F(x),M(x))=0$, and write ε_2 in terms of ε_1, we obtain the same asymptotic expansion as the curve C_h^{loc}, which shows the consistency of the analysis of global Hopf bifurcation obtained in Section 4.2 with analysis of local subcritical Hopf bifurcation derived from the Bogdanov–Takens bifurcation.

By relations $\varepsilon_1=\alpha-\alpha^*$ and $\varepsilon_2=\mu-\mu^*$, we are able to sketch the curves of saddle-node bifurcation C_{sn}, Hopf bifurcation C_h^{loc} and homoclinic bifurcation C_{hom} in a small neighborhood of the point $BT(\alpha^*,\mu^*)$. Note that we have analytic expression $\Delta(\alpha,\mu)=0$ for the curve of saddle-node bifurcation C_{sn} and $\Delta_2(\alpha,\mu,x_1(\alpha,\mu))=0$ for the curve of Hopf bifurcation C_h, so we are able to extend the curves of saddle-node bifurcation and Hopf bifurcation in a neighborhood of (α^*,μ^*) to the boundary of region D_2. For the curve of homoclinic bifurcation C_{hom}, we

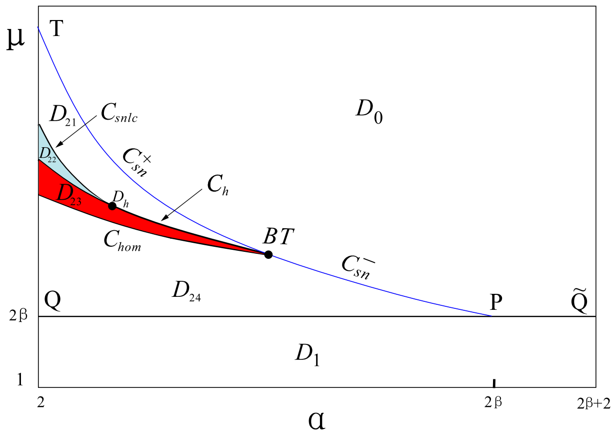
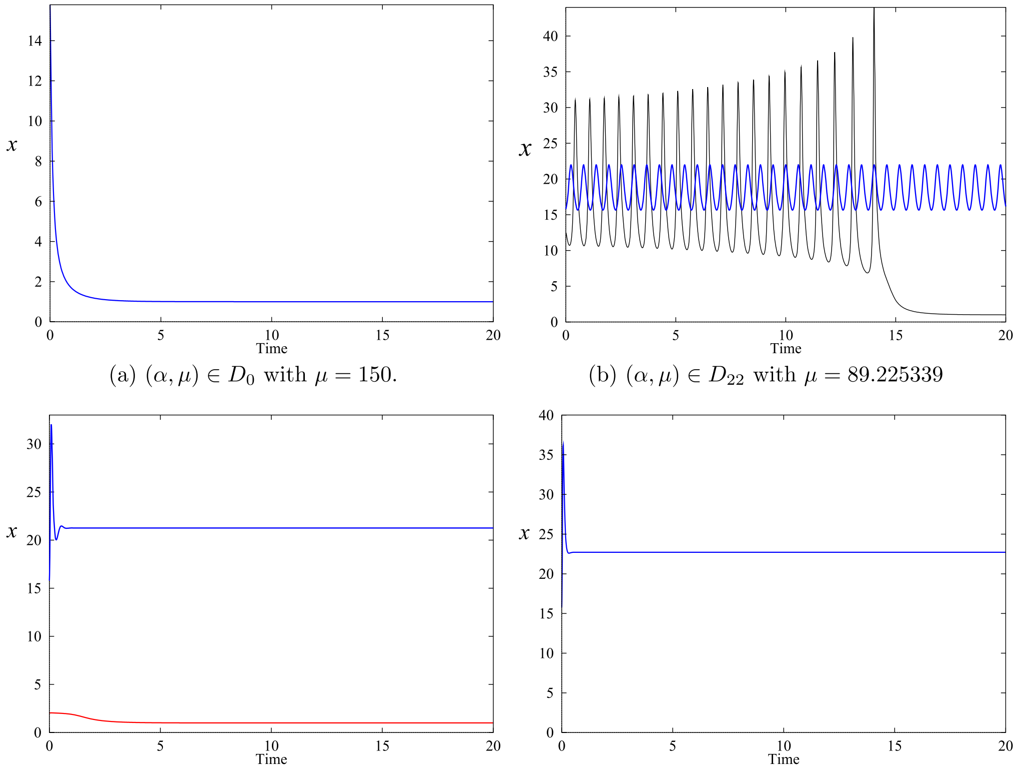


Fig. 6. Bifurcation diagrams and regions of multiple limit cycles for $\beta > 3 + \sqrt{6}$. There is a semi-stable and an unstable limit cycle on the curve C_{snlc} and the region D_{23}, respectively. In the curved triangular region D_{22}, there are two limit cycles surrounding the unstable equilibrium E_1, the inner one is stable and the outer one is unstable.

plot it near the point $BT(\alpha^*, \mu^*)$ by expression in the definition of C_{hom}, and extend the curve to the boundary of D_2 with the help of simulation. According to the analysis in Section 4.2, on the curve of Hopf bifurcation there is a unique point D_h at which the first Lyapunov coefficient l_1 vanishes, and the codimension of Hopf bifurcation is at least two. The arc between the points D_h and BT are C_h^+ across which subcritical Hopf bifurcation occurs, and the curve C_h^{loc} near the point BT is a subarc of C_h^+. The supcritical Hopf bifurcation occurs on the arc C_h^-. For the degenerate Hopf bifurcation, there is a curve of saddle-node bifurcation of limit cycles C_{snlc}, through which multiple limit cycles can be generated in a small neighborhood of the point D_h. See Fig. 4 and simulations in Section 4.2. We roughly sketch the curve C_{snlc} with the help of simulation and extend it to the boundary of region D_2.

Hence, we obtain a bifurcation diagram of system (2.2) for $\beta > 3 + \sqrt{6}$. See Fig. 6. The half infinite region D is divided into six regions, D_0, D_1, D_{2i} $(i = 1, 2, 3, 4)$, where $D_2 = \cup_{i=1}^{i=4} D_{2i}$. We go over these regions counterclockwise from D_0, in which the unique equilibrium E_0 is locally asymptotically stable. An unstable node E_1 and a saddle point E_2 appear through a saddle-node bifurcation, when the ordered pair (α, μ) crosses the curve C_{sn}^+ and enters the region D_{21}. System (2.2) undergoes a saddle-node bifurcation of limit cycles when (α, μ) crossing the arc C_{snlc}, on which there is a semi-stable limit cycle surrounding the unstable focus E_1. This semi-stable limit cycle bifurcates into two hyperbolic limit cycles in the curved triangular region D_{22}, the inner one is stable and the outer one is unstable. If (α, μ) passes through the curve C_h^- into the curved triangular region D_{23}, equilibrium E_1 gains the stability and meanwhile the inner limit cycle disappears by the supercritical Hopf bifurcation. There is a unique, unstable limit cycle in the region D_{23}. This unstable limit cycle disappears from either the homoclinic bifurcation when (α, μ) crosses the curve C_{hom} into the region D_{24}, or subcritical Hopf bifurcation when (α, μ) crosses the curve C_h^+ back into the region D_{21}. When (α, μ) crosses the line segment PQ into the region D_1, the saddle point E_2 disappears through a backward bifurcation, and there is a unique, stable endemic equilibrium E_1 in region D_1, which disappears via a forward bifurcation when (α, μ) goes back to the region D_0 through line segment $P\tilde{Q}$.

Simulation. Fix $\alpha = 2.1$ and $\beta = 11.711918 > 3 + \sqrt{6}$, and choose different values of μ such that (α, μ) is in different subregions of D. We plot the graph of $x = x(t)$. The



(a) $(\alpha, \mu) \in D_0$ with $\mu = 150$.

(b) $(\alpha, \mu) \in D_{22}$ with $\mu = 89.225339$

(c) $(\alpha, \mu) \in D_{24}$ with $\mu = 45$.

(d) $(\alpha, \mu) \in D_1$ with $\mu = 15$.

Fig. 7. Time series of x for (α, μ) in regions D_0, D_{22}, D_{24} and D_1 where $\beta = 11.711918$.

three initial conditions are used in the simulation. (I1): $(15.78636, 0.08274288, 0.241556)$; (I2) $(12.5, 0.08274288, 0.241556)$; (I3) $(2, 0.08274288, 0.241556)$.

(a) $(\alpha, \mu) \in D_0$ with $\mu = 150$. The solution with initial condition (I1) converges to the unique equilibrium $E_0(1, 0, 1)$. See Fig. 7(a).
(b) $(\alpha, \mu) \in D_{22}$ with $\mu = 89.225339$. The solution with initial condition (I1) converges to the stable limit cycle. If we choose another set of initial condition (I2), the corresponding solution converges to the stable disease-free equilibrium E_0. See Fig. 7(b).
(c) $(\alpha, \mu) \in D_{24}$ with $\mu = 45$. The solution with initial condition (I1) converges to the stable endemic equilibrium E_1, while the solution with initial condition (I3) converges to the stable disease-free equilibrium E_0. See Fig. 7(c).
(d) $(\alpha, \mu) \in D_1$ with $\mu = 15$. The solution with initial condition (I1) converges to the stable endemic equilibrium E_1. See Fig. 7(d). In this case, E_0 is unstable.

Compared to the classical epidemic model on networks, our model on adaptive networks shares the common feature if β is small: the basic reproduction number R_0 characterizes the dynamics of the epidemic model. Disease will spread if $R_0 > 1$ and be eliminated otherwise. However, if β is sufficiently large, our adaptive-network model is significantly different from the classical network-epidemic model and presents complex dynamics in the following two aspects:

• *Bistability.* Our model exhibits two types of bistability: (a) coexistence of two stable equilibria E_0 and E_1 in region $D_{23} \cup D_{24}$; (b) coexistence of stable equilibrium E_0 and a stable limit cycle in region D_{22}. The phenomenon of bistability indicates that the disease transmission is initial-dependent. Whether a disease will outbreak depends on the initial populations of all compartments. For given initial populations, the basins of attraction of the equilibria E_0, E_1 and/or the stable limit cycle will determine the trend of the diseases.

• *Multiple periodic solutions.* Our model exhibits two types of oscillating behaviors: (a) transient oscillations; (b) sustained oscillations. The oscillation phenomenon accounts for the recurrence of some infectious diseases. In the region D_{23}, system (2.2) has a unique unstable limit cycles. If the initial condition is sufficiently close to this unstable limit cycles, we can observe the transient oscillation and disease will either die out or outbreak eventually. In the region D_{22} system (2.2) has multiple limit cycles, one of which is stable. Disease will present cyclic pattern in the long run if the initial conditions lie in the basin of attraction of the stable limit cycles; otherwise, the disease will eventually die out.

5. Conclusion and discussion

The study of the mechanism and dynamics of epidemic is important to understand and take measures to control the transmission of infectious diseases. How to characterize the mulstability and recurrence of many infectious diseases has been attracted tremendous interests of epidemiologists. A large amount of studies of classical well mixing epidemic models have shown that the nonlinear incidence rate and recovery rate play a key role in generating these complicated dynamics [17,24–26,32]. On the other hand, epidemic modeling on networks has greatly progressed in the understanding of the interplay between network properties and contagion processes. Gross et al. [11] employed the pairwise epidemic models [13,14] on complex networks to investigate the interaction between the change of networks' topological structure and epidemic dynamics. Similar dynamics arising in classical well mixing epidemic system including periodic oscillations are found in this network epidemic model numerically.

In order to study the impact of the topology of adaptive networks caused by rewiring mechanism on the disease transmission, in this paper we provide a complete analysis of the dynamical behaviors for the SIS pairwise epidemic model on adaptive networks proposed by Gross et al. [11]. We have shown that the model has very rich and complicated dynamics analytically, including backward bifurcation, saddle-node bifurcation, Hopf bifurcation and Bogdanov–Takens bifurcation of codimension 2. The rewiring rate w (or μ) plays an important role in determining whether or not the epidemic outbreaks. From the study in Section 3, it is known that rewiring rate w (or μ) determines the basic reproduction number R_0 and the stability of the equilibria. Recalling that

$$R_0 = \frac{2\beta}{\mu} = \frac{\tau \langle k \rangle}{\gamma + w},$$

the larger the rewire rate is, the smaller the disease propagation threshold R_0 is. The impact of rewiring rate w on the epidemic dynamics is illustrated in the bifurcation diagrams of Figs. 1, 5 and 6. We have the conclusion as following.

(1) When $\tau \langle k \rangle \leqslant \gamma$ (i.e., $2\beta \leqslant 1$), the model has a unique stable disease-free equilibrium, and the disease will die out since R_0 is always less than 1 regardless of the value of w. Therefore rewiring rate w will not create rich dynamics.

(2) When $\gamma < \tau\langle k\rangle \leqslant \tau + 2\gamma$ (i.e., $1 < 2\beta \leqslant \alpha$), a typical forward bifurcation takes place at $R_0 = 1$. The dynamics of the model are characterized by R_0. The disease will be eliminated when $R_0 < 1$; otherwise, the disease will persist. Hence, the disease can be controlled by increasing rewiring rate such that R_0 is below 1.

(3) In contrast, when $\tau\langle k\rangle > \tau + 2\gamma$ (i.e., $2\beta > \alpha$), rewiring rate w may dramatically change the dynamics of the model. If there is no rewiring or the rewiring rate w is very small, R_0 is greater than 1 and the disease will outbreak. If the rewiring rate w is very large such that R_0 is less than the subthreshold R_0^c, the disease will be eliminated. However, if rewiring rate w is intermediate such that $R_0^c < R_0 < 1$, the existence of Hopf bifurcation of codimension 2 and subcritical Bogdanov–Takens bifurcation implies that the disease may exhibits bistability, transient oscillations or sustained oscillations depending the other two parameters τ and γ. See region D_2 in Fig. 6. Therefore, rewiring rate w will change the disease transmission in many counterintuitive ways in this case.

In addition, we continues to think more about the underlying reasons of these rich dynamics. The evolving behavior of the infection over time is represented as $\dot{I} = \beta SI - \gamma I$ for classical well mixing SIS models, and this equation turns into $\dot{I} = \beta wSI - \gamma I$ if the rewiring mechanism is incorporated. Considering the infection force item of βwSI, it can be easily found that the rewire rate changes the disease transmission ability. However, it is not enough characterize the full dynamics of the diseases from our study in Section 4. Hence, the conjunction of adaptive human behaviors and the complex structure of the network has significant impacts on the epidemic transmission, both of which are indispensable in generating the complicate dynamics.

Acknowledgments

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Appendix A. Affine maps in Theorem 3.4

If $a_1^2(x_c) - 4a_2(x_c) > 0$, we apply the affine map

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} + \begin{bmatrix} \beta(\bar{\mu} - \alpha + 1) & 1 & 1 \\ 1 & \frac{4(\theta_1+1)}{\beta(\alpha-2\beta-2)^2} & \frac{4(\theta_2+1)}{\beta(\alpha-2\beta-2)^2} \\ -\frac{\bar{\mu}}{2} & \frac{4(2\beta-\alpha)(\theta_1+1+2\beta)}{\beta(\alpha-2\beta+2)(\alpha-2\beta-2)^2} & \frac{4(2\beta-\alpha)(\theta_2+1+2\beta)}{\beta(\alpha-2\beta+2)(\alpha-2\beta-2)^2} \end{bmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}.$$

If $a_1^2(x_c) - 4a_2(x_c) = 0$, we apply the affine map

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} + \begin{bmatrix} \beta(\bar{\mu} - \alpha + 1) & 1 & 0 \\ 1 & \frac{2(\alpha-4\beta+1)}{\beta(\alpha-2\beta-2)^2} & \frac{1}{\beta(\bar{\mu}-\alpha+1)} \\ -\frac{\bar{\mu}}{2} & \frac{2(2\beta-\alpha)(\alpha+1)}{\beta(\alpha-2\beta+2)(\alpha-2\beta-2)^2} & \frac{4(2\beta-\alpha)}{\beta(\alpha-2\beta+2)(\alpha-2\beta-2)^2} \end{bmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}.$$

If $a_1^2(x_c) - 4a_2(x_c) < 0$, we apply the affine map

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_c \\ y_c \\ z_c \end{pmatrix} + \begin{bmatrix} \beta(\bar{\mu}-\alpha+1) & 0 & 1 \\ 1 & \frac{2\sqrt{4a_2(x_c)-a_1^2(x_c)}}{\beta(\alpha-2\beta-2)^2} & \frac{2(\alpha-4\beta+1)}{\beta(\alpha-2\beta-2)^2} \\ -\frac{\bar{\mu}}{2} & \frac{2(2\beta-\alpha)\sqrt{4a_2(x_c)-a_1^2(x_c)}}{\beta(\alpha-2\beta+2)(\alpha-2\beta-2)^2} & \frac{2(2\beta-\alpha)(\alpha+1)}{\beta(\alpha-2\beta+2)(\alpha-2\beta-2)^2} \end{bmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix}.$$

Appendix B. Proof of Theorem 4.2

Taking the affine transformation (4.4) into (4.8) yields the system

$$\begin{cases} \dot{u} = v + l_{00010}\varepsilon_1 + l_{00001}\varepsilon_2 + L_{20}u^2 + L_{11}uv + L_{02}v^2 \\ \qquad + \sum\limits_{\substack{i+j+k+p+q\geqslant 2 \\ i+j\neq 2 \text{ if } i+j+k+p+q=2}} l_{ijkpq}u^i v^j w^k \varepsilon_1^p \varepsilon_2^q, \\ \dot{v} = m_{00010}\varepsilon_1 + m_{00001}\varepsilon_2 + M_{20}u^2 + M_{11}uv + M_{02}v^2 \\ \qquad + \sum\limits_{\substack{i+j+k+p+q\geqslant 2 \\ i+j\neq 2 \text{ if } i+j+k+p+q=2}} m_{ijkpq}u^i v^j w^k \varepsilon_1^p \varepsilon_2^q, \\ \dot{w} = \lambda_3 w + g_{00010}\varepsilon_1 + g_{00001}\varepsilon_2 + \sum\limits_{i+j+k+p+q\geqslant 2} g_{ijkpq}u^i v^j w^k \varepsilon_1^p \varepsilon_2^q, \end{cases} \tag{B.1}$$

where

$$l_{00010} = \frac{2(x_c-1)(x_c-2)\left(2x_c^2-5x_c+4\right)}{\lambda_3^2(x_c-3)}, \qquad l_{10010} = -\frac{\left(2x_c^2-5x_c+4\right)x_c}{\lambda_3^2},$$
$$l_{00001} = \frac{2(x_c-1)(x_c-2)\left(3x_c^2-6x_c+4\right)}{x_c\lambda_3^2(x_c-3)}, \qquad l_{10001} = -\frac{3x_c^2-6x_c+4}{\lambda_3^2},$$
$$l_{10100} = \frac{(x_c-3)\left(2x_c^7-21x_c^6+89x_c^5-210x_c^4+302x_c^3-272x_c^2+148x_c-40\right)x_c}{-2\lambda_3^2},$$
$$l_{01100} = \frac{(x_c-3)\left(4x_c^7-42x_c^6+183x_c^5-446x_c^4+654x_c^3-568x_c^2+264x_c-48\right)x_c}{4\lambda_3^2},$$
$$l_{00200} = \frac{(x_c-3)\left(6x_c^5-33x_c^4+78x_c^3-94x_c^2+56x_c-12\right)(x_c-2)^3x_c}{8\lambda_3^2},$$
$$l_{00110} = -\frac{x_c(x_c-1)\left(2x_c^2-5x_c+4\right)(x_c-2)^2}{2\lambda_3^2},$$
$$l_{00101} = -\frac{(x_c-1)\left(3x_c^2-6x_c+4\right)(x_c-2)^2}{2\lambda_3^2},$$
$$m_{00010} = -\frac{4(x_c-2)(x_c-1)^2}{x_c(x_c-3)\lambda_3}, \qquad m_{00001} = -\frac{4(x_c-1)(x_c-2)}{x_c(x_c-3)\lambda_3},$$
$$m_{10100} = \frac{x_c(x_c-3)\left(x_c^2-3x_c+3\right)\left(x_c^3-6x_c^2+8x_c-4\right)}{\lambda_3}, \qquad m_{10010} = \frac{2(x_c-1)}{\lambda_3},$$
$$m_{01100} = -\frac{x_c(x_c-3)\left(2x_c^5-18x_c^4+61x_c^3-106x_c^2+94x_c-32\right)}{2\lambda_3}, \qquad m_{10001} = 2\lambda_3^{-1},$$
$$m_{00200} = -\frac{(x_c-3)\left(3x_c^2-7x_c+6\right)(x_c-2)^4x_c}{4\lambda_3}, \qquad m_{00110} = \frac{(x_c-1)^2(x_c-2)^2}{\lambda_3},$$
$$m_{00101} = \frac{(x_c-1)(x_c-2)^2}{\lambda_3}, \quad g_{00010} = \frac{4(x_c-2)(x_c-1)^2}{x_c\lambda_3^2(x_c-3)}, \qquad g_{00001} = \frac{4(x_c-1)^2}{\lambda_3^2(x_c-3)}.$$

The local center manifold takes the form

$$w = -\frac{g_{00010}}{\lambda_3}\varepsilon_1 - \frac{g_{00001}}{\lambda_3}\varepsilon_2 + \sum_{i+j+p+q\geqslant 2} h_{ijpq}u^i v^j \varepsilon_1^p \varepsilon_2^q,$$

where the coefficients h_{ijpq} are omitted due to complexity. System (B.1) restricted to the center manifold is

$$\begin{cases} \dot{u} = v + l_{00010}\varepsilon_1 + l_{00001}\varepsilon_2 + L_{20}u^2 + L_{11}uv + L_{02}v^2 + \sum\limits_{\substack{i+j+p+q\geqslant 2 \\ i+j\neq 2 \text{ if } i+j+p+q=2}} l_{ijpq}u^i v^j \varepsilon_1^p \varepsilon_2^q, \\ \dot{v} = m_{00010}\varepsilon_1 + m_{00001}\varepsilon_2 + M_{20}u^2 + M_{11}uv + M_{02}v^2 + \sum\limits_{\substack{i+j+p+q\geqslant 2 \\ i+j\neq 2 \text{ if } i+j+p+q=2}} m_{ijpq}u^i v^j \varepsilon_1^p \varepsilon_2^q, \end{cases}$$

where

$$\begin{aligned}
&l_{1010} = l_{10010} - \lambda_3^{-1} g_{00010}\, l_{10100}, \quad l_{1001} = l_{10001} - \lambda_3^{-1} g_{00001}\, l_{10100}, \\
&l_{0110} = -\lambda_3^{-1} g_{00010}\, l_{01100}, \quad l_{0101} = -\lambda_3^{-1} g_{00001}\, l_{01100}, \\
&l_{0020} = \lambda_3^{-2} g_{00010}\, (g_{00010}\, l_{00200} - \lambda_3\, l_{00110}), \\
&l_{0002} = \lambda_3^{-2} g_{00001}\, (g_{00001}\, l_{00200} - \lambda_3\, l_{00101}), \\
&l_{0011} = \lambda_3^{-2}\, (2\, g_{00001}\, g_{00010}\, l_{00200} - \lambda_3\, g_{00001}\, l_{00110} - \lambda_3\, g_{00010}\, l_{00101}), \\
&m_{1010} = m_{10010} - \lambda_3^{-1} g_{00010}\, m_{10100}, \quad m_{1001} = m_{10001} - \lambda_3^{-1} g_{00001}\, m_{10100}, \\
&m_{0110} = -\lambda_3^{-1} g_{00010}\, m_{01100}, \quad m_{0101} = -\lambda_3^{-1} g_{00001}\, m_{01100}, \\
&m_{0020} = \lambda_3^{-2} g_{00010}\, (g_{00010}\, m_{00200} - \lambda_3\, m_{00110}), \\
&m_{0002} = \lambda_3^{-2} g_{00001}\, (g_{00001}\, m_{00200} - \lambda_3\, m_{00101}), \\
&m_{0011} = \lambda_3^{-2}\, (2\, g_{00001}\, g_{00010}\, m_{00200} - \lambda_3\, g_{00001}\, m_{00110} - \lambda_3\, g_{00010}\, m_{00101}).
\end{aligned}$$

Using the near-identity transformation

$$\begin{aligned}
u =& (1 + L_{1010}\varepsilon_1 + L_{1001}\varepsilon_2 + \mathcal{O}(\varepsilon^2))X + (L_{0010}\varepsilon_1 + L_{0001}\varepsilon_2 + \mathcal{O}(\varepsilon^2)) + (\mathcal{O}(\varepsilon^2))Y \\
&+ \frac{1}{2}(L_{11} + M_{02} + \mathcal{O}(\varepsilon))\, X^2 + (L_{02} + \mathcal{O}(\varepsilon))XY + (\mathcal{O}(\varepsilon))Y^2 + \mathcal{O}\left(|X, Y|^3\right), \\
v =& (1 + M_{0110}\varepsilon_1 + M_{0101}\varepsilon_2 + \mathcal{O}(\varepsilon^2))Y - (l_{00010}\varepsilon_1 + l_{00001}\varepsilon_2 + \mathcal{O}(\varepsilon^2)) + (M_{1010}\varepsilon_1 + M_{1001}\varepsilon_2 \\
&+ \mathcal{O}(\varepsilon^2))X - (L_{20} + \mathcal{O}(\varepsilon))X^2 + (M_{02} + \mathcal{O}(\varepsilon))XY + (\mathcal{O}(\varepsilon))Y^2 + \mathcal{O}\left(|X, Y|^3\right),
\end{aligned}$$

where

$$
\begin{aligned}
L_{0010} &= \frac{M_{11}\, l_{00010}+M_{02}\, m_{00010}-m_{1010}}{2M_{20}}, \ L_{0001} = \frac{M_{11}\, l_{00001}+M_{02}\, m_{00001}-m_{1001}}{2M_{20}},\\
L_{1010} &= \frac{15\, x_c^9-131\, x_c^8+486\, x_c^7-988\, x_c^6+1174\, x_c^5-784\, x_c^4+221\, x_c^3+46\, x_c^2-50\, x_c+12}{\lambda_3^3 x_c (x_c-1)},\\
L_{1001} &= \frac{60\, x_c^8-528\, x_c^7+1987\, x_c^6-4144\, x_c^5+5182\, x_c^4-3916\, x_c^3+1704\, x_c^2-376\, x_c+32}{4\lambda_3^3 (x_c-1)^2},\\
M_{1010} &= -\frac{\left(x_c^2-2\right)\left(3\, x_c^2-10\, x_c+6\right)}{\lambda_3^2 x_c}, \ M_{1001} = \frac{(x_c-2)\left(x_c^3-10\, x_c^2+18\, x_c-8\right)}{\lambda_3^2 (x_c-1)},\\
M_{0110} &= \frac{5\, x_c^7-38\, x_c^6+114\, x_c^5-170\, x_c^4+121\, x_c^3-18\, x_c^2-25\, x_c+10}{\lambda_3^2 x_c (x_c-1)},\\
M_{0101} &= \frac{20\, x_c^6-160\, x_c^5+515\, x_c^4-864\, x_c^3+804\, x_c^2-396\, x_c+80}{4\lambda_3^2 (x_c-1)^2},
\end{aligned}
$$

rewriting X, Y into u, v and regrouping terms, we obtain

$$
\begin{cases}
\dot{u} = v,\\
\dot{v} = \overline{\nu}_1(\varepsilon) + \overline{\nu}_2(\varepsilon)v + (M_{20} + \mathcal{O}(\varepsilon^2))u^2 + (2L_{20} + M_{11} + \mathcal{O}(\varepsilon))uv + v\mathcal{O}\left(|u, v|^2\right),
\end{cases}
\tag{B.2}
$$

where

$$
\begin{aligned}
\overline{\nu}_1 &= -\frac{4(x_c-2)(x_c-1)^2}{x_c\,(x_c-3)\lambda_3}\varepsilon_1 - \frac{4(x_c-1)(x_c-2)}{x_c\,(x_c-3)\lambda_3}\varepsilon_2 - \frac{x_c-2}{\lambda_3 x_c (x_c-3)}\varepsilon_1\varepsilon_2 + \mathcal{O}(\varepsilon^3),\\
\overline{\nu}_2 &= -\frac{16\, x_c^5-102\, x_c^4+239\, x_c^3-256\, x_c^2+124\, x_c-24}{\lambda_3^2}\varepsilon_1 - \frac{16\, x_c^5-103\, x_c^4+243\, x_c^3-262\, x_c^2+128\, x_c-24}{\lambda_3^2 (x_c-1)}\varepsilon_2 + \mathcal{O}(\varepsilon^2).
\end{aligned}
$$

Finally by rescaling the variables and time, we change system (B.2) to system (4.9) with

$$
\nu_1(\varepsilon) = \frac{(2L_{20} + M_{11} + \mathcal{O}(\varepsilon))^4}{(M_{20} + \mathcal{O}(\varepsilon^2))^3}\overline{\nu}_1(\varepsilon), \quad \nu_2(\varepsilon) = \frac{2L_{20} + M_{11} + \mathcal{O}(\varepsilon)}{M_{20} + \mathcal{O}(\varepsilon^2)}\overline{\nu}_2(\varepsilon).
$$

We can represent $\nu_1(\varepsilon)$ and $\nu_2(\varepsilon)$ in the form (4.10), and have

$$
\left|\frac{\partial\,(\nu_1,\ \nu_2)}{\partial\,(\varepsilon_1,\ \varepsilon_2)}\right|_{\varepsilon=0} = \frac{128\, x_c^3\,(x_c-2)\left(x_c^2-3\, x_c+3\right)^5}{\lambda_3^8\,(x_c-1)^3}.
$$

Substituting expressions of λ_3 and x_c in the formula above, the desired non-degeneracy property is obtained.

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